

## FALCON's Response to Delhivery's RFQ for Bhiwandi Warehouse Automation for Parcel Sortation

December 9, 2022



[www.falconautotech.com](http://www.falconautotech.com)

**Kind Attention –**

**Mr. Sunny Raja  
Mr. Nikhil Ummat**

**M/s Delhivery Ltd**

**Offer Ref: F22-00409; Date 09.12.2022**

**Subject – Techno-Commercial Offer for Warehouse Automation\_Shipment Sorter  
(Bhiwandi Location).**

Dear Team,

We are pleased to submit our Techno-commercial Offer in response to your RFQ of parcel sorter for your Bhiwandi mega hub.

As you will note, we have studied your requirement in great depth and, along with the information collected during calls with you, we have put together a detailed techno-commercial proposal laid out in various sections and sequenced to enable you to understand our proposal and to re-enforce our commitment to be your partner in this important project.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Sortation Technologies and references.

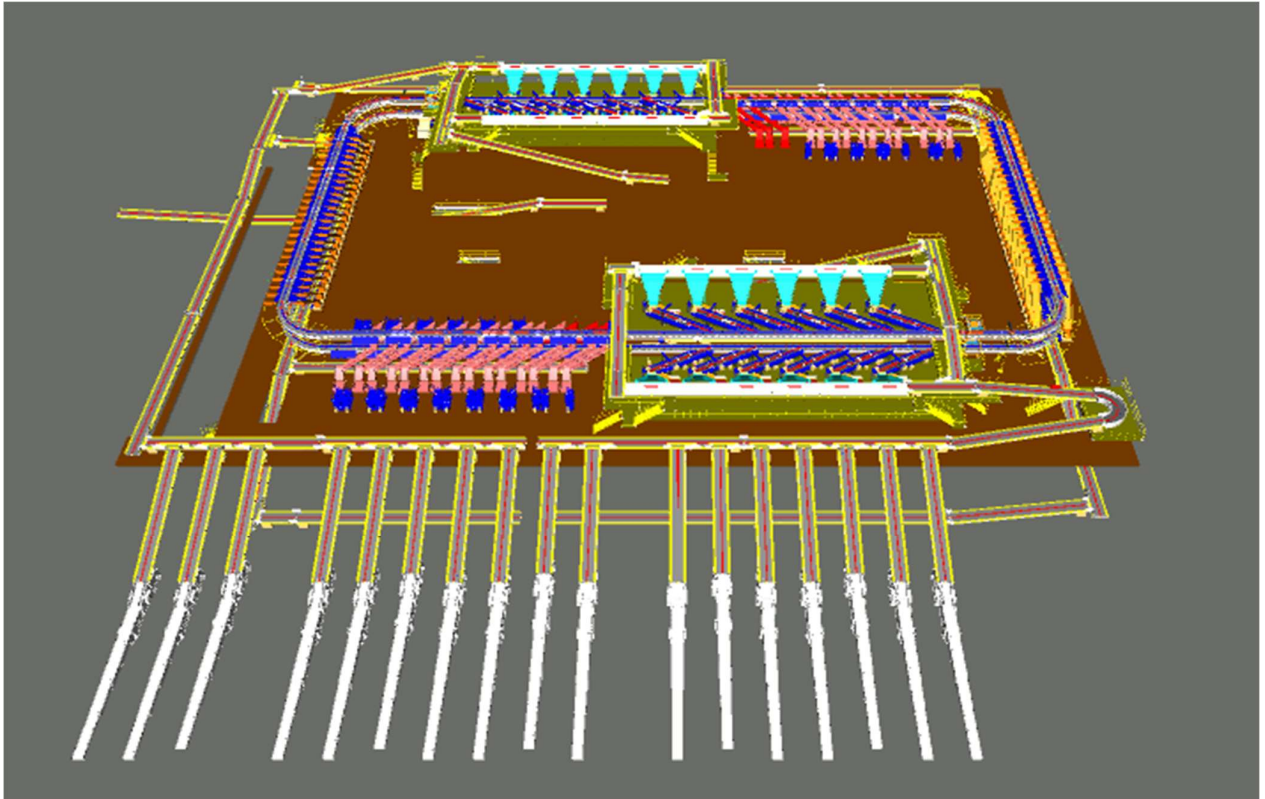
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through your award process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

**Best Regards,**

**Adib Hasim**

**Manager – Sales & Solution**

## Double Deck Cross Belt Sorter ~ 32K TPH



**Proposal Reference – F22-00409**

**Falcon Autotech Private Limited  
Plot No 308 & 309  
Ecotech-1 Extension, Greater Noida  
Uttar Pradesh- 201308, India.**

**Contact – Adib Hasim  
Mob - +91 9711 619998  
Adib.hasim@falconautoonline.com**

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## 1. Glossary

| S. No. | Term | Description                       |
|--------|------|-----------------------------------|
| 1      | RFQ  | Request For Quote                 |
| 2      | PPH  | Parcels Per Hour                  |
| 3      | DBO  | Damaged Barcode                   |
| 4      | ICR  | Intelligent Character Recognition |
| 5      | MEZZ | Mezzanine                         |
| 6      | LIM  | Linear Induction Motors           |
| 7      | ECDS | Empty Carrier Detection System    |
| 8      | AC   | Alternating Current               |
| 9      | DC   | Direct Current                    |
| 10     | PLC  | Programmable Logic Controller     |
| 11     | IT   | Information Technology            |
| 12     | BOQ  | Bill Of Quantity                  |
| 13     | I/O  | Input/ Output                     |
| 14     | PDP  | Power Distribution Panel          |
| 15     | PC   | Personal Computer                 |
| 16     | UPS  | Uninterrupted Power Supply        |
| 17     | CBS  | Cross Belt Sorter                 |
| 18     | DNF  | Data Not Found                    |
| 19     | LGM  | Logic Mismatch                    |
| 20     | RTVC | Real Time Video Coding            |
| 21     | VDS  | Volume Distribution System        |

## 2. Company Profile

### THE COMPANY

#### ABOUT US

Falcon Autotech is one of the leading Intra-logistics automation solution providers with a sharp focus on Parcel, Piece and Case Handling applications. We have a broad portfolio of Sortation Systems for Retail, E-Commerce and CEP (Courier, Express and Postal) industries and have supplied our systems across various countries globally.

#### OUR VISION

To create a global impact on the society by facilitating world's leading organizations run their supply chains efficiently by leveraging technology enabled products and solutions and, in the process, establish Falcon Autotech as one of the world's most respected and trusted brands in Intra-logistics warehouse automation space.

#### OUR JOURNEY

The team started out in 2004 solving special purpose automation problems for clients and later pivoted to building a standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems. Over the last few years, the organization has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

#### OUR CORE COMPETENCIES

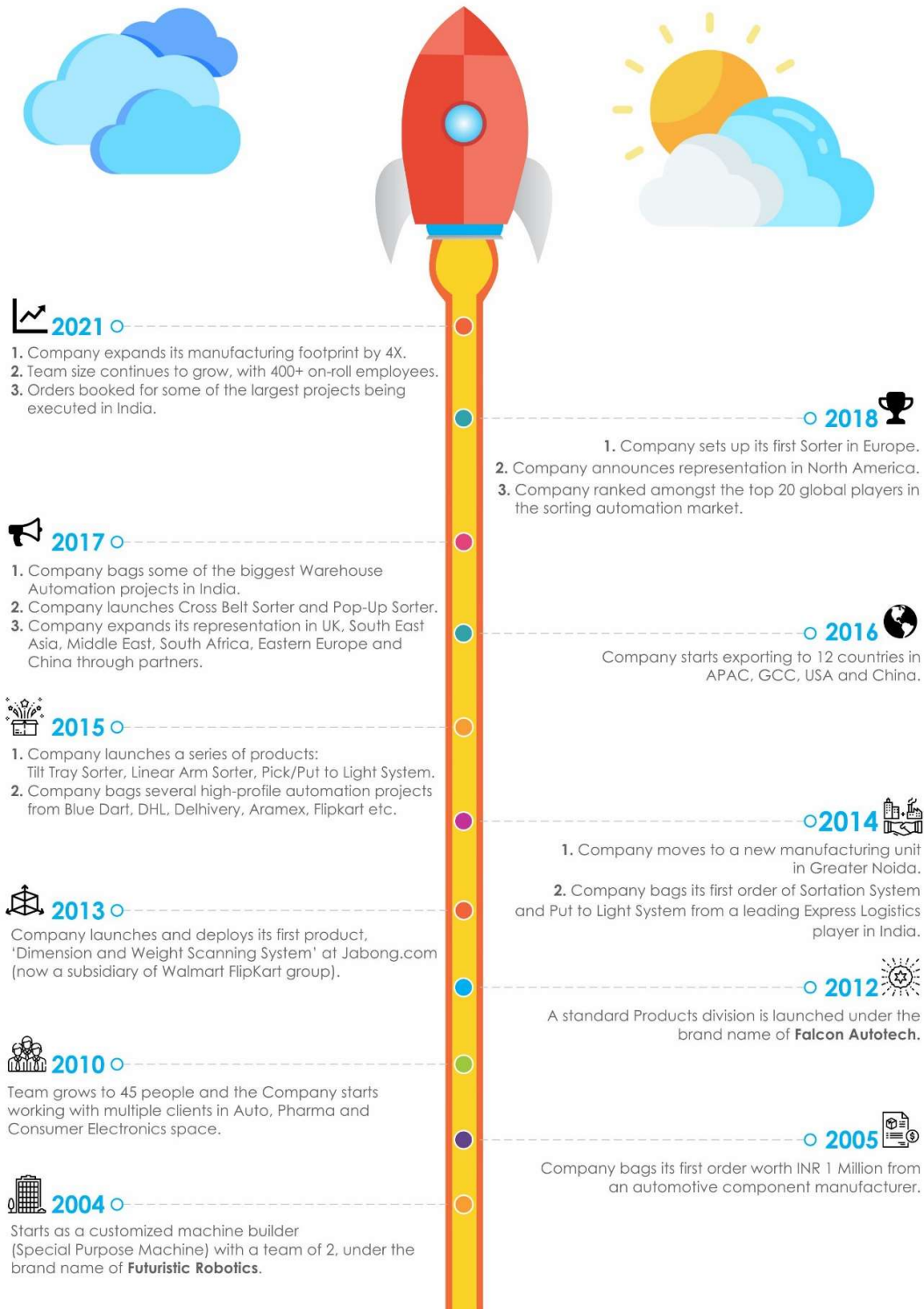
Organizations are built and defined by the people they are able to attract, cultivate and retain and the talent management philosophy at Falcon Autotech could not agree more. At, Falcon Autotech we take a lot of pride in being the preferred career choice for some of the finest talent in the country across multiple domains such as:

- ❖ Supply Chain Solution Designing and Consultation
- ❖ Hardware Design
- ❖ Electrical Design
- ❖ Electronics Design
- ❖ Software Development
- ❖ Manufacturing

Giving us an unprecedented ability to harness our true potential generated by culmination of Supply Chain Know-How, quality Hardware and flexible Software under one roof.



## COMPANY HISTORY



## KEY STATISTICS

---

**04**Product Lines

---

**400+**On-Roll Employees

---

**215+**Engineering Team

---

**~150,000 Sq Ft**Manufacturing Space

---

**20+**Tier A Clients

---

**15+**Countries with Live Installations

---

**1700+**Total Installed Systems Globally

---

**35**Representations in 4 Continents and 35 Countries

---

**1 Million+**Sorts Per Hour of combined Sorting Capacity  
installed Globally

---



## OUR PRODUCT PORTFOLIO

01

### SORTATION SOLUTIONS



Cross Belt Sorter



Tilt Tray Sorter



Arm Sorter



Swivel Wheel Sorter



Sweep Sorter



Pusher Sorter



Pop Up Sorter

02

### PICK/PUT TO LIGHT SYSTEMS



Packet Sortation



Multi-Line Order Consolidation



Order Picking



Store Order Fulfillment



Spare Parts Distribution



Kitting



Multi Order Pick Cart

03

### DIMENSION & WEIGHT SCANNING SYSTEMS

#### STATIC DIMENSION & WEIGHT SCANNING SYSTEMS



Cubizon-R



Cubizon-R THRU



Cubizon-R Cross



Cubizon-R ECO

#### DYNAMIC CONVEYOR BASED SYSTEMS

##### OUR 8 MODELS



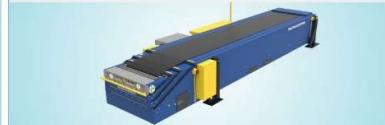


# OUR PRODUCT PORTFOLIO

04

## CONVEYOR AUTOMATION SOLUTIONS

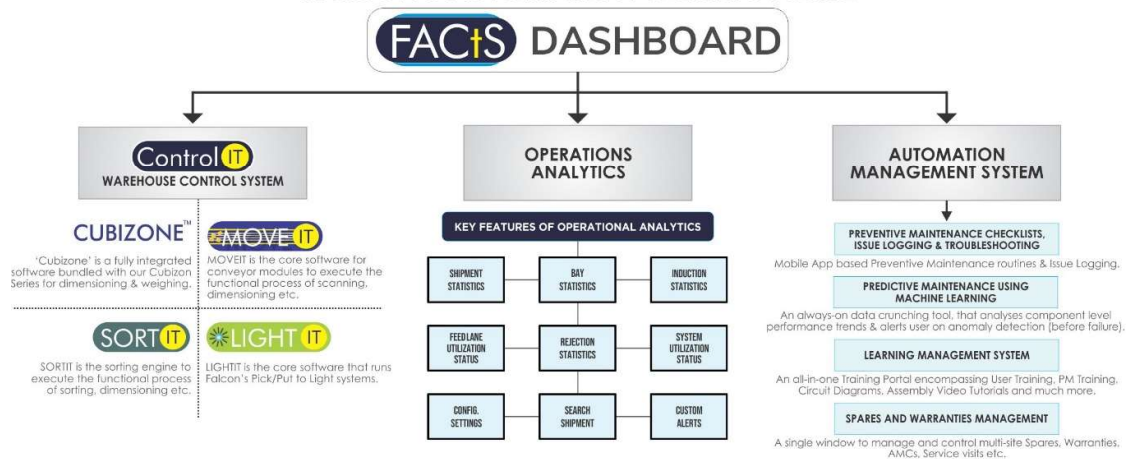
| BELT CONVEYORS                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ROLLER CONVEYORS                                                                                                                                                                                                                                                                                                         | SPECIAL APPLICATION SOLUTIONS                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>❖ <b>PVC BELT</b> <ul style="list-style-type: none"> <li>◆ Straight Conveyor</li> <li>◆ Spacing Conveyor</li> <li>◆ Accumulation Conveyor</li> <li>◆ Strip Belt Merge- 30/45/60</li> <li>◆ Solid Belt Merge- 45</li> <li>◆ Incline Belt Conveyor</li> </ul> </li> <li>❖ <b>MODULAR BELT</b> <ul style="list-style-type: none"> <li>◆ Straight Conveyor</li> <li>◆ 90 Degree Belt Curve</li> <li>◆ 180 Degree Belt Curve</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>◆ Straight Conveyor</li> <li>◆ Gravity Roller Conveyor</li> <li>◆ MDR Accumulation</li> <li>◆ Angled Roller Conveyor</li> <li>◆ Curved Conveyor</li> <li>◆ Spur Merge Conveyor</li> <li>◆ Swivel Wheel Diverter</li> <li>◆ Pop Up Belt Diverter</li> <li>◆ Pusher Unit</li> </ul> | <ul style="list-style-type: none"> <li>◆ Aligning Conveyor</li> <li>◆ One to Two Split Conveyor</li> <li>◆ Two to One Merge Module</li> <li>◆ Bulk Distribution-ARB</li> <li>◆ Bulk Distribution-Telescopic</li> <li>◆ Bulk Distribution-Electric Arm</li> <li>◆ Telescopic Belt Conveyor</li> </ul> |



05

## SOFTWARE SUITE

### FALCON'S AUTOMATION CONTROL SUITE



### — ADD-ON MODULES —



#### Real Time Video Coding

No-Read Barcodes have always been a challenge in Sorter Applications globally. With Falcon's RTVC feature, we can now easily decode such challenging parcels in real time while sitting remotely.



#### Deep Learning Image Classification

DLIC is an artificial intelligence function that imitates the working of the human brain. It can identify objects such as packets on the carrier and classify them in different categories.



#### Cloud Sort App/Virtual PTL

It's an android based app used to perform secondary sorting in light assisted put-away operations. It can work as a stand-alone setup or can be integrated with any sorter.

## SOME OF OUR KEY PROJECTS

Falcon Autotech has pioneered a variety of unique Warehouse Automation solutions and has been successfully running several automation projects such as, high-speed Sortation Systems, static and online Weighing & Dimensioning Systems, Pick/Put to Light Systems, complex Conveying Systems etc. for E-Commerce, Courier, Express and Postal, FMCG, Cosmetics, Food, Pharma, Automobile, and Logistics Industries.

01



A high-speed Cross Belt Sortation project with one of India's leading E-commerce players for their high-volume site (>500k sq. ft. area). The solution is designed with a Volume Distribution System for handling the bulk volumes, Integrated Pre-Sorters, 28 automatic inducts with automatic weighing, barcode scanning, dimensioning and image capture capabilities. Falcon's Loop Cross Belt Sorter is designed to sort 27,600 parcels per hour and sort directly into 410 bagging chutes. All bags are closed sealed, scanned and taken away with the help of takeaway conveyors.

02



A complete end-to-end automation solution designed for a Pickup Processing Centre including Automated Shipment Unloading and Scanning, Automatic Conveying of shipments, Automatic Shipment Sortation at 24000 parcels per hour with automatic DWS, sorting into a total of 1080 end destinations with the help of primary and secondary sorting (using PTLs). This system is designed with a unique, proprietary feature provided by Falcon Autotech to handle the irregular parcels on the machine. This solution also consists of Bag Conveying, Automatic Bag Sortation at 3600 parcels per hour, sorting into 38 routes and Automated Loading of bags/shipments into the vehicles.

03



A high-speed Cross Belt Sortation solution designed for India's Largest E-commerce player for one of their high-volume sites. This solution caters to a throughput of 24000 parcels per hour, sorting directly into 100 end destinations. This solution is also designed with Falcon's Conveyor Automation modules for efficient material movement.

04



A complete end-to-end warehouse automation solution with one of India's largest Fashion Retail companies. The solution comprises of Automatic Unloading, Automatic Goods Receipt & Reconciliation, Put-away, Picking and Packing automation delivered through Conveyor automation modules and Vertical Reciprocating Conveyors. It also comprises of Falcon's high-speed Cross Belt Sortation Systems delivering a throughput of 16000 parcels per hour, for primary sortation combined with Put to Light System for secondary sortation.



## INDUSTRY SOLUTIONS

As a leading player in the intra-logistics automation space, Falcon Autotech continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.

### FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling

### FOCUS INDUSTRIES



E-COMMERCE RETAIL



COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



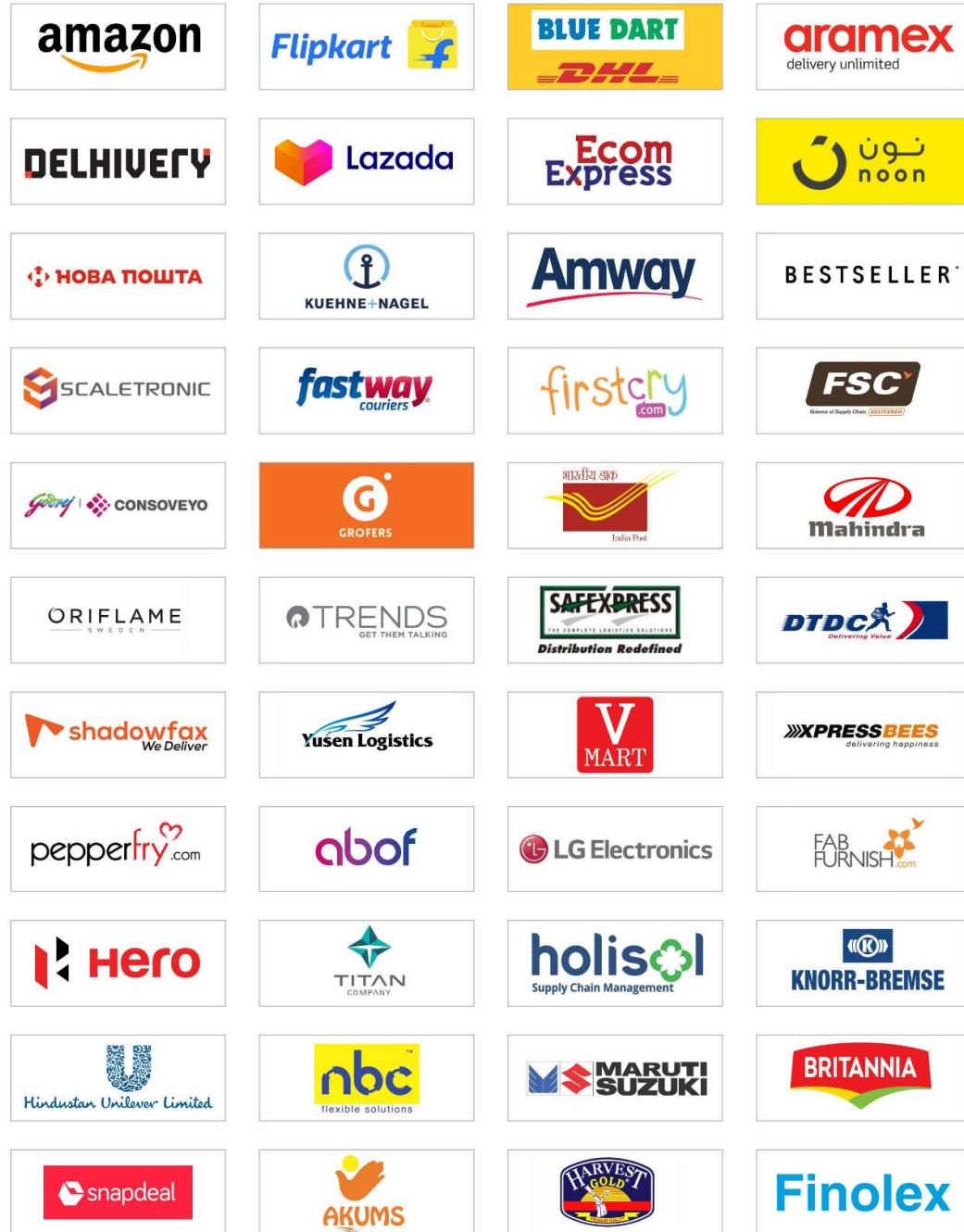
SPARE PARTS DISTRIBUTION



GROCERY DISTRIBUTION

## SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.



## MEDIA COVERAGE



Global Logistics Automation Market\* report published by Data Bridge Market Research, has featured Falcon Autotech amongst the **top 20 in global list of sortation systems manufacturers**. CNBC Awaaz, Awaaz Entrepreneur, featured Falcon Autotech as the **Fastest Growing Logistics Automation company** in India. Besides being featured in many important daily newspapers, we have also won the **Best Warehouse Automation Company of the Year** award at Express Logistics Supply Chain Conclave held in Mumbai in the Month of September 2015.

**BusinessLine**  
Home News BUDGET 2017 Markets Companies Money & Banking Economy Info  
Social Media Computers & Laptops Mobiles & Tablets Other Gadgets

### Robotics powers festival sale impact on productivity for e-commerce brands

JESSE JOHN

October 26, 2016: Between Amazon's 'Great Indian Festival', Snapdeal's 'Utsav Divali' and Flipkart's 'Big Billion Day' sales, all eyes were on deals and wins earlier this month. However, what's common for most brands expecting large impact sales seasons is the need for operational efficiency. Falcon Autotech, an industrial automation solution provider in India, enables efficient fulfilment in the face of a surge of orders experienced by many online retailers during these high demand periods.

Vinay Bail, CEO, Falcon Autotech, explains, "It's the warehouses or fulfilment centres that see 2 to 5.5 times the regular volumes for large festive sale periods."

**EFFICIENCY UNPLUGGED**

**THE FINANCIAL EXPRESS**

### How automation firm Falcon helping clients deliver products to workers

Industrial automation firm Falcon is helping its clients use robotic carts to pick, place and deliver products from specific locations to workers

Ever wondered how an e-commerce company fulfills thousands of orders everyday—picking up the product from the warehouse nearest the delivery address, packing it properly, labelling it with the right barcode and mandatory receipts, and then ensuring that it reaches the customer in minimal time? While an army of men and machines work round the clock to make it happen,

**The Economic Times**  
**SmallBiz**

### Falcon Autotech looks overseas for growth

By Priyanka Jangam, ET Bureau | Updated Nov 01, 2016, 08:34 AM IST

PUNE: Falcon Autotech, a Noida-based maker of warehousing automation products, expects international operations to account for about a third of its revenue over the next few years, even as it sees demand for its

**THE ASIAN AGE**  
Dhaka | Mumbai | Kolkata | London

### Hidden technologies changing our shopping experience

E-commerce sites finally solve the conundrum of sustainable growth and profitability

**Are consumers**  
E-commerce sites have been trying to gain traction for years. But the real challenge is to ensure the same level of efficiency in the warehouse as in the storefront. Falcon Autotech, an industrial automation solution provider in India, enables efficient fulfilment in the face of a surge of orders experienced by many online retailers during these high demand periods.

**Are retailers**  
It is a given that the last product in the supply chain is the one that matters the most. The warehouse is the final frontier for e-commerce. Falcon Autotech, an industrial automation solution provider in India, enables efficient fulfilment in the face of a surge of orders experienced by many online retailers during these high demand periods.

**DECCAN Chronicle**

### These tech oriented companies are making it smooth for industries to run

TECHNOLOGY IN OTHER NEWS

Industries still need to cross a lot of hurdles. Technology is playing role in every beat and is innovating to resolve the challenges. Here are few organizations that have made it better for industries in India. Not only have they saved time but are also saving a chunk of money for different industries.

Aaveg manages the entire activity chain including Employee transportation, Leasing to business rental and advising, managing and procuring the overall

**PRODUCT CARGOTALK**

### Cutting edge tech for dimensioning & weighing

The highly accurate and versatile Cubicon-R THRU is equipped with roller bed base, has robust operations along with guaranteed zero-maintenance.

Cubicon-R THRU helps a company with warehouse storage space planning and shipment freight costs

Cubicon-R THRU is a highly accurate and versatile roller bed base, has robust operations along with guaranteed zero-maintenance.

\*Source: Report by Data Bridge Market Research



## CLIENT TESTIMONIALS

“ Working with Falcon Autotech has been a very delightful experience for us, BLUE DART being a leader in the express logistics business, we expect the same kind of responsiveness, reliability, commitment & passion from our partners and Falcon has always delivered to their promise on all the solutions such as the Auto Sorter, Weight Dimension Labelling machine, Parcel Locker etc. All the technology interventions have been business critical for us.



Mr. Anil Khanna  
Managing Director, DHL BLUE DART

“ Falcon is an organisation that has shown great agility and has helped us improve our service capabilities. Their sortation systems have helped us deal effectively with our growing volumes. We have found the Falcon team quite responsive to feedback and quick to incorporate changes into their systems that have helped build better, customized solutions for Delhivery.



Mr. Sahil Barua  
CEO, DELHIVERY

“ Falcon Autotech has been efficient and proactive, particularly in meeting deadlines and their technical advancements. Falcon Autotech strike just the right balance between being a flexible, easy to deal with firm yet having all the key strengths and attributes. Every member of the team is honest, reliable, responsive, personable and very knowledgeable. I wish them good luck for all the future endeavours.



Late Mr. Sanjeev Saxena  
Director, ECOM EXPRESS

“ I believe you guys earned it with your hard work and dedication. I believe the products you have designed demonstrate innovation and lateral thinking. Also the management has a flexible and the customer centric approach. It has been a delight to work with you and the team.



Mr. Amitava Saha  
CEO, firstcry.com

### 3. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers** globally. Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolio in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 Million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.

**Born2Invest** **Incas-SSI Schaefer merger creates leading player in Italy's intralogistics market**

Incass, the transportation management segment of the logistics automation industry is expected to have a higher CAGR. This is due to factors associated with the growing need to deliver goods in a timely manner and to reduce transportation costs.

Moreover, automotive will be the leading segment of the industry from 2018 to 2023, with an increasing demand for automated production systems and flexible logistics systems used in manufacturing and supply of vehicle fleets.

The MarkestandMarkets research report has identified SSI Schaefer as one of the key players in the logistics automation market worldwide during the forecast period, along with Dematic Corporation (US), Murata Machinery, Ltd. (Japan), Jungheinrich AG (Germany), and Falcon Autotech (India), among others.

This year's largest market for logistics automation is estimated to be North America, according to the research report. Regardless, Incas and SSI Schaefer's merger could help them compete in their main market.

(Featured image by DepositPhotos)

0 SHARES

Not secure | worldindustryinsights.com/30269/global-parcel-sortation-system-market-analysis-size-share-growth-trends-and-forecast-2019-2026-siemens-vanderlande-beumer-group

## Global Parcel Sortation System Market Analysis, Size, Share, Growth, Trends, and Forecast 2019-2026: Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives

BY ANN.CASTRO ON AUGUST 12, 2019

The global "Parcel Sortation System market" research report is a pervasive research report that touches the most vital parts of the Parcel Sortation System market that is necessary to be grasped by a professional or even a layman. The market research report enlightens one regarding few of the important aspects such as an overview of the Parcel Sortation System product, the growth factors enhancing or hampering its development, application in the various fields, major dominating companies, genuine facts, economic situation, and geographical analysis. The research report endows the data concerning the dynamics that propel the growth as well as the supply and demand chain of the product on a global basis.

Click here to access the report:: [www.reportsbuzz.com/request-for-sample.html?repid=50822](http://www.reportsbuzz.com/request-for-sample.html?repid=50822)

The international and global stand of the Parcel Sortation System market is also briefly mentioned in the research report based on the performed statistical and thorough market analysis. The information mentioned in the research report gives a qualitative and quantitative view of the overall market. The statistical analysis of the market helps analyze the supply, demand, production, maintenance, and storage costs of the product. The data regarding some of the prevalent players Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives, Dematic, Interroll, Muratec, Invata Intralogistics, Viastore, Okura Yusoki, GBI Intralogistics, Equinox Mhe, Pitney Bowes, Falcon Autotech, OCM SRL, Solystic, Bowe Systec, Intralox are also detailed in the current case study.

## 4. Reference projects

Falcon has a strong legacy in **Warehousing Automation** solutions and takes pride in-

1. Expertise in delivering high throughput Sortation systems for bags and boxes.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

### 4.1 Project 1- (Ecommerce Client)

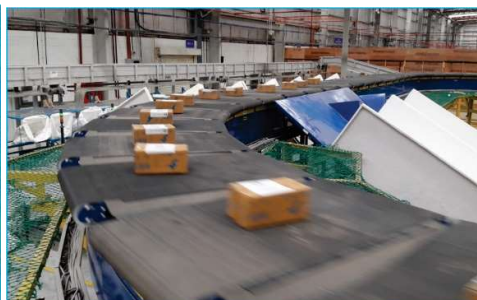
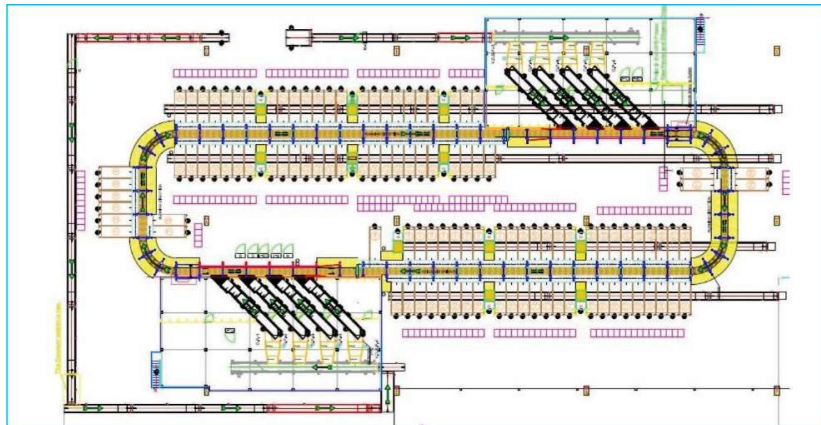
**Use Case** - Destination Sorting of Packed Parcels

**Solution Specifications** –

- Throughput: 16,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

**Key Technology Modules** - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

**Layout and Site Pictures** -





## 4.2 Project 2- (Client – Courier Express Parcel)

**Use Case** - Courier Sorting of Parcels

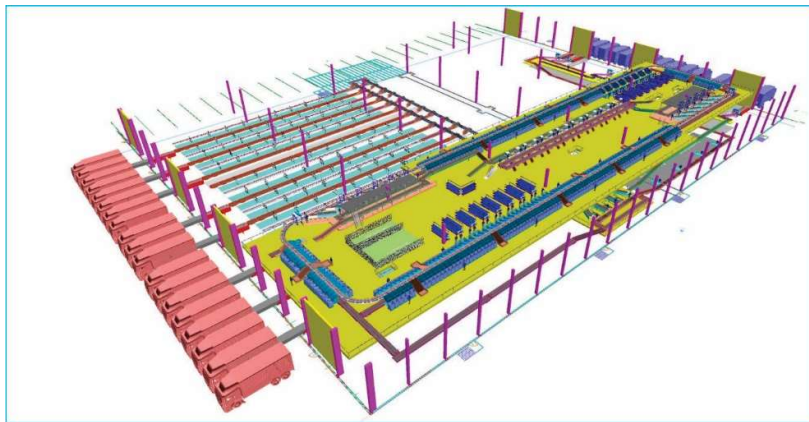
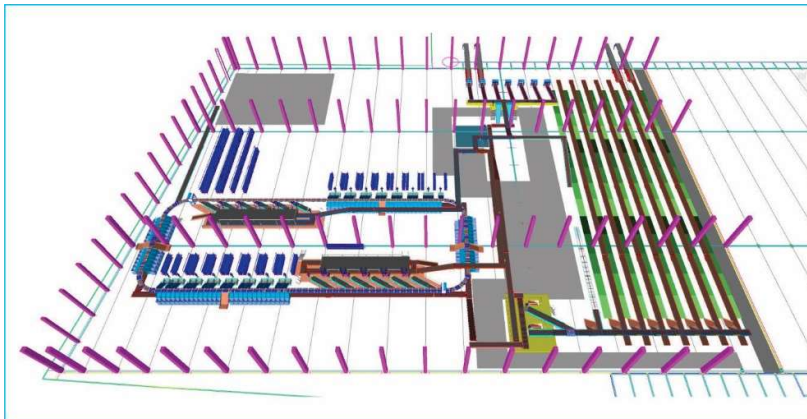
**Solution Specifications** –

- Throughput: 24,000 PPH
- End Destinations: 394 Direct, 30 Secondary, 630 PTLs
- Building Size: 200,000 Sq Ft

**Key Technology Modules** –

- Pre-Sorting
- Telescopic Belt Conveyors
- Bag Sorters
- PTL System

**Layout** -



**Layout & Pictures**



### 4.3 Project 3- (Client – E-Commerce)

**Use Case** – Destination Sorting of Packed Parcels.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C parcels. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

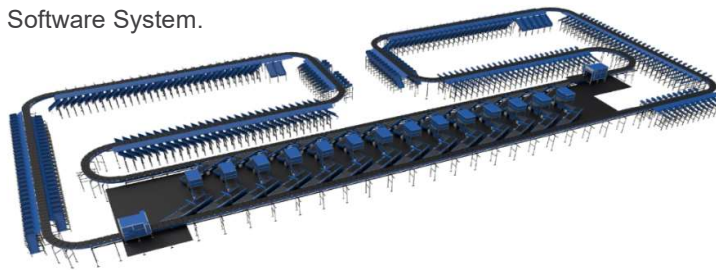
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

#### **Solution Specifications –**

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

#### **Key Technology Modules –**

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.



#### **Layout & Pictures**



#### 4.4 Project 4- (CEP Client)

In 2019, customer selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing of parcels arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

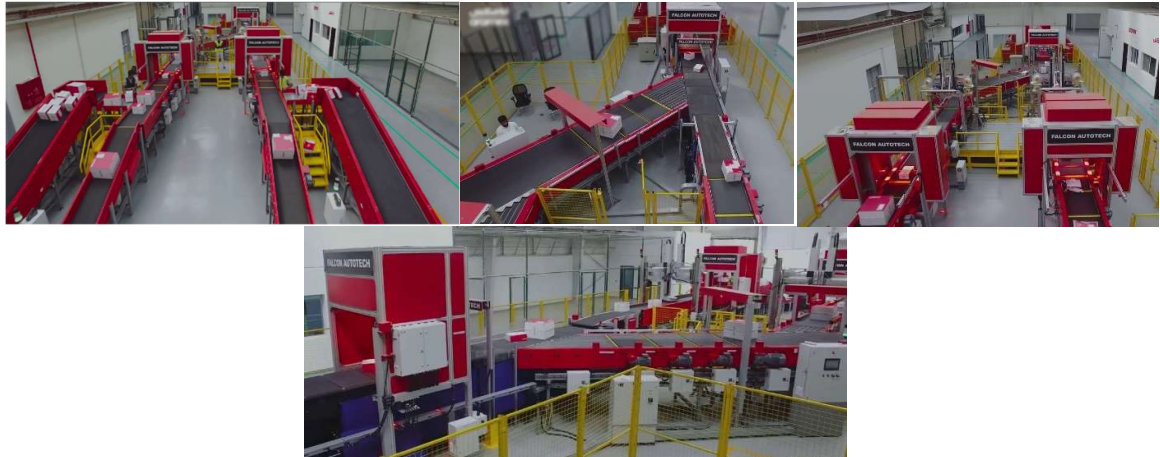
##### **Solution Specifications –**

- Throughput: 4800 PPH
- End Destinations: 52 Nos

##### **Key Technology Modules –**

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Parcel handling.
- FOCR Engine.
- WCS Software System.

##### **Site Picture –**



## 5. Handled Parcel Spectrum for your Solution

### 5.1 Parcel size loadable on the sorter

| Parameter           | Unit | Minimum                                                              | Maximum |
|---------------------|------|----------------------------------------------------------------------|---------|
| Length              | mm   | 100                                                                  | 600     |
| Width               | mm   | 50                                                                   | 400     |
| Height              | mm   | 5                                                                    | 300     |
| Weight              | Kg   | 0.05                                                                 | 10      |
| Min Height measured | mm   | 20                                                                   |         |
| Dimension Accuracy  | mm   | (+/-10)                                                              |         |
| Weight Accuracy     | Kg   | +/- 60 grams or 0.05% of the Weight of Product (whichever is higher) |         |

### 5.2 Parcels to be loaded on Sorter shall have the following characteristics:

1. Centre of Gravity of Item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behaviour of parcel cannot be guaranteed.
3. Parcels shall be perfectly and safely packed: protrusion or open surfaces are not allowed.
4. Plastic ropes shall be perfectly adherent to the surface of the package.
5. Items must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
6. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
7. Items shall not have slippery surfaces and must be able to withstand acceleration of the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
8. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
9. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.  
All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

### 5.3 Parcel not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
2. Items that are packed in material that can damage the conveyors or the sorter.

3. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
4. Items that have been classified as dangerous.
5. Wet items.
6. Items with anti-slip treatment.
7. Items with protruding parts.
8. Inadequately packed items that could be damaged during automatic transportation.
9. Electrostatically loaded items.
10. Loose parts on Items such as adhesive tape, stickers, slips of paper, straps, wrap foil etc.
11. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint.
12. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system

## 5.4 Pictures of Sortable Parcels

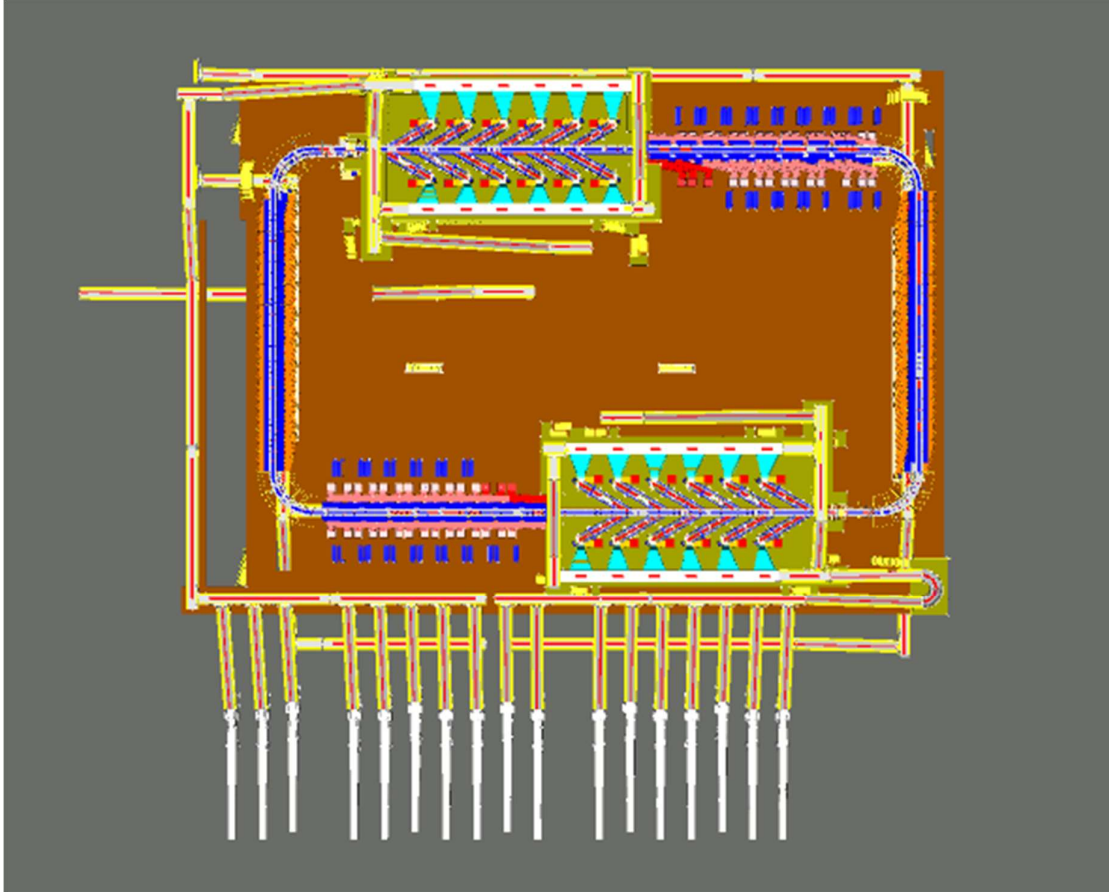
| PARCEL TYPE                                                                         |                                                    | DESCRIPTION                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Strongly Strapped                                  | A parcel banded with strapping material.                                                                                                                                                                              |
|   | Pre Manufactured Plastic Courier Pack <A3 in size  | A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.                                                                                                                        |
|  | Pre Manufactured Plastic Courier Pack > A3 in size | A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed. |
|  | Other Plastic Covered Item                         | A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.                                                                                                                 |
|  | Paper Covered Item                                 | A cardboard parcel coated in a packing paper.                                                                                                                                                                         |
|  | Cloth Covered Item                                 | Often Import parcels that are of irregular shape.                                                                                                                                                                     |
|  | Wine Box                                           | A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.                                                                                        |
|  | Cardboard Box                                      | A cardboard parcel is often the norm.                                                                                                                                                                                 |
|  | Totes                                              | A large plastic, lidded box.                                                                                                                                                                                          |
|  | Flats                                              | A parcel that often contains documents.                                                                                                                                                                               |
|  | Poly Bags                                          | It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.                                                                                                      |



## 6. Proposed System Description

### 6.1 Objective

The purpose of this proposal is the design, manufacturing, installation, commissioning, testing and acceptance testing of a complete Sortation system to handle Parcels .



**Proposed System Layout**

### 6.2 Summary of the System

- **Key silent features of the solution:**
  - o Double deck loop cross belt sorter to be installed on the mezz....
  - o Common Chute for both upper and lower deck...

- Interconnected VDS loop for upper and lower deck...
- Safety devices are considered as per Falcon Standards and referenced from previous projects

- **Process Flow:**

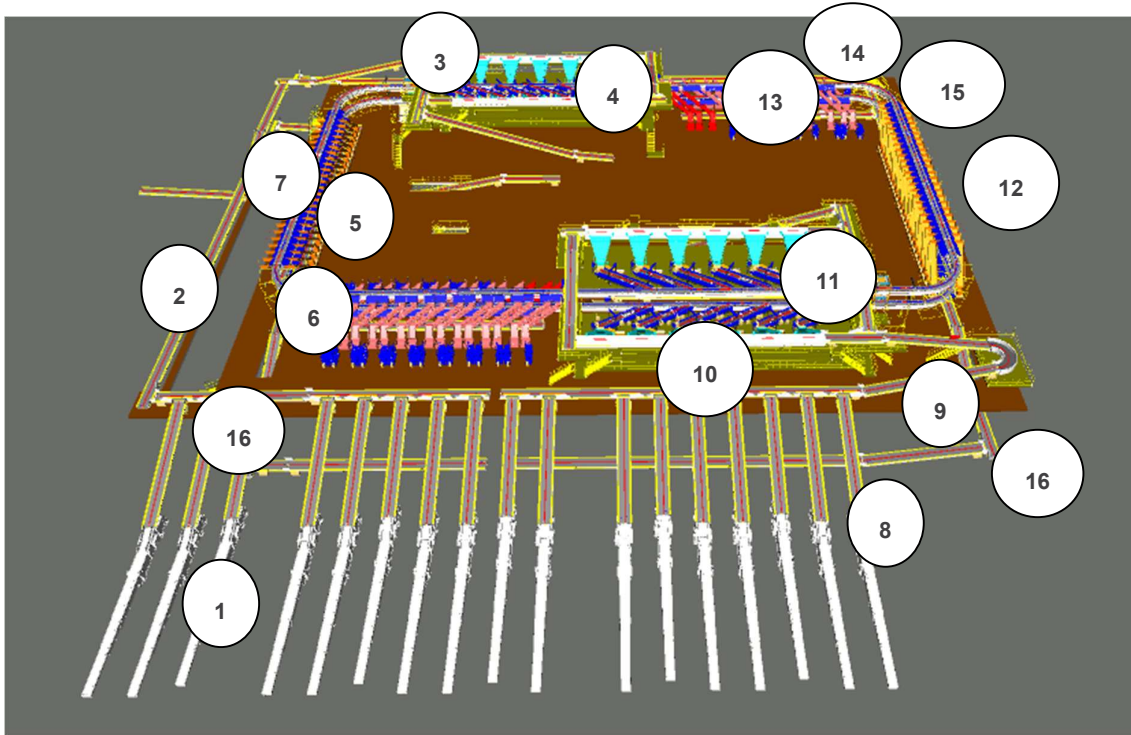
- Shipments will be unloaded from truck on the Telescopic Belt Conveyor.
- TBC will move the shipments to the respective infeed highway conveyors via inclined conveyors
- In case of oversize parcel, operator will remove the parcels from the TBC and put this on Oversize Parcel Conveyors, which will transfer the big shipments directly to the VDS Loop of Bag Sorter
- Infeed Highway Conveyors will move the shipments to VDS Loop Conveyors
- ARB based Volume Distribution Systems (VDS) distribute the shipments to the different inducts
- The VDS transfers the shipments to Accumulation Chutes which lead up to the induct lines.
- Operators orient and place the shipments on the Induct lines equipped with sensors
- Feedlines induct the parcels on the Sorter automatically onto the Cross-Belt Sorter Loop which then sorts the parcels into respective chute using data from Flipkart's Sorting decision system. The Dimensions and image of the parcel is also captured on the Sorter.
- The sorted shipments are discharged into Direct Bagging Chutes and PTL Chutes.
- Secondary Sorting happens at PTL Chutes with help of Put to Light Racks and Bags are prepared. Associates scan the Shipments and drop them into the Bags guided by Light Modules on Racks.
- Take out Conveyors are planned near the chutes. Once the Bags are filled, they are dropped onto the Take-out Conveyors

### 6.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput of ~32k parcels per hour.
2. Ability to handle entire range of conveyable Parcels with a sorting conveyor speed up to 2.1 m/s.
3. Low occupancy of floor space in the building using efficient Double Deck Loop CBS layout and chute design.
4. Narrow discharge centres for increased number of splits in limited space.
5. FALCON's CBS can adapt with changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear.

## 7. Layout View



### Legend

- 1 - Zone-1 Infeed Telescopic Conveyor – 8 nos.
- 2- Zone-1 Infeed Highway Conveyors
- 3- Zone-1 VDS
- 4- Zone-1 Feedlines – 12 Nos ( 6 Nos on each Deck)
- 5 – Zone-1 Direct Bagging Chutes
- 6 – Zone-1 PTL Chutes
- 7 – Zone-1 Bag takeout Conveyors
- 8 – Zone-2 Infeed Telescopic Conveyor – 9 nos.
- 9- Zone – 2 Infeed Highway Conveyors
- 10- Zone-2 VDS
- 11- Zone-2 Feedlines – 12 Nos (6 Nos on each Deck)
- 12 – Zone-2 Direct Bagging Chutes
- 13– Zone-2 PTL Chutes
- 14 – Zone-2 Bag takeout Conveyors
- 15- Main Loop, Double Deck Cross Belt Sorter
- 16- Big Parcel Handling Conveyor



## 8. Proposed System Capacity\_ Calculations

### 8.1 Sorter System Capacity

The following table shows the Throughput Calculation for the Sorter

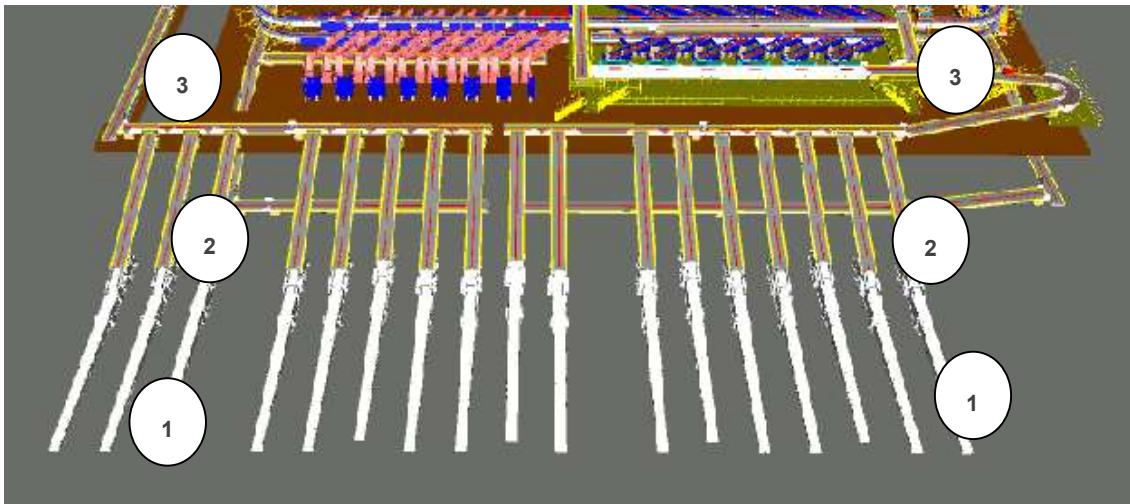
|                                                |       |     |
|------------------------------------------------|-------|-----|
| Sorter Speed                                   | 2.1   | m/s |
| Carrier Pitch                                  | 0.6   | m   |
| Carriers per Hour/ Zone                        | 12600 | CPH |
| Sorter Utilization Factor                      | 1.33  |     |
| Sorter Throughput (1 Deck)(Design Capacity)    | 16758 | PPH |
| Sorter Throughput (Both Deck)(Design Capacity) | 33516 | PPH |

$$\text{Carriers Per Hour/Zone} = (\text{Sorter Speed} * 3600) / (\text{Carrier Pitch})$$

$$\text{Sorter TPH} = \text{Carrier Per Hour/Zone} * \text{Sorter Utilization Factor}$$

## 9. System description

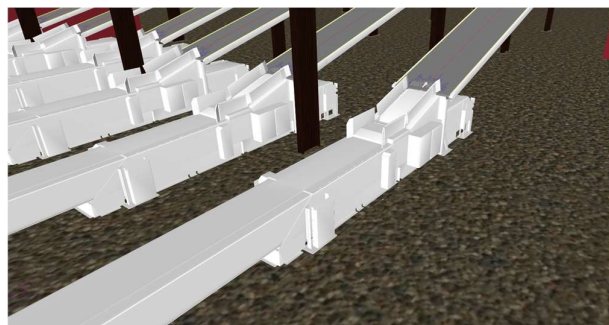
### 9.1 Infeed System



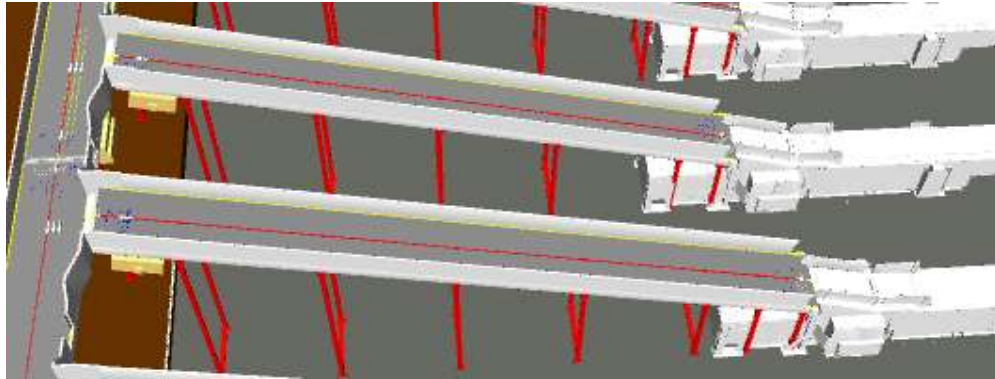
1. Telescopic Belt Conveyor
2. Inclined Conveyor
3. Highway Conveyor

#### 9.1.1 Telescopic belt Conveyor

Operator will unload the shipments from the truck in bulk on Telescopic Conveyor.



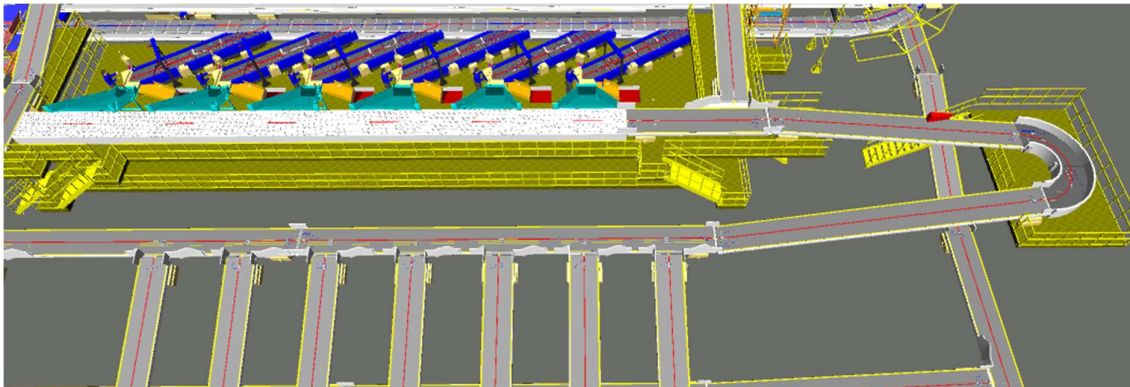
### 9.1.2 Inclined Conveyors



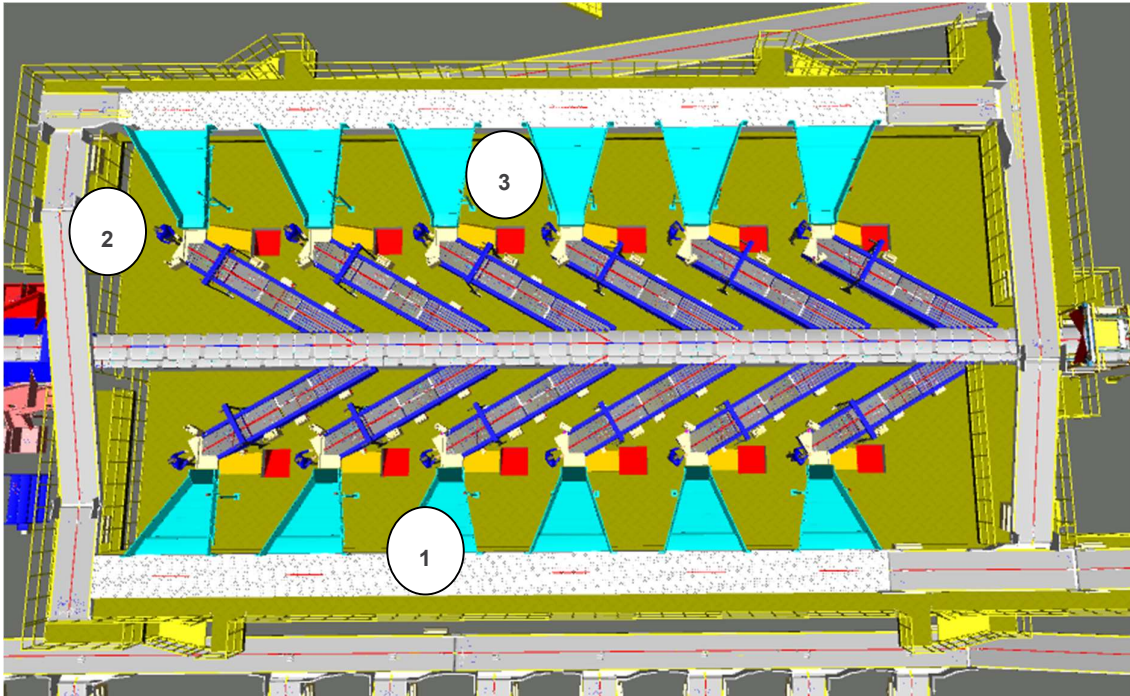
Shipment will move from TBC to the inclined conveyors. Inclined Conveyors will transfer the load on infeed highway conveyors.

### 9.1.3 Infeed Highway Conveyors

Infeed Highway Conveyors will transfer the load from inclined conveyors to the VDS Loop Conveyor. Infeed Highway Conveyors are belt type conveyors

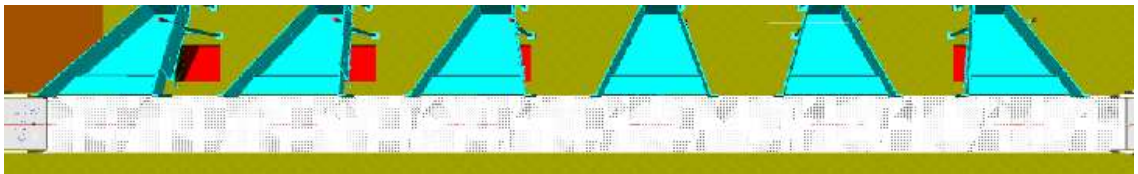


## 9.2VDS System

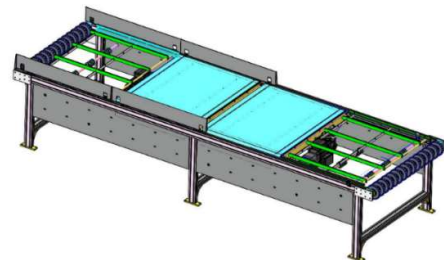
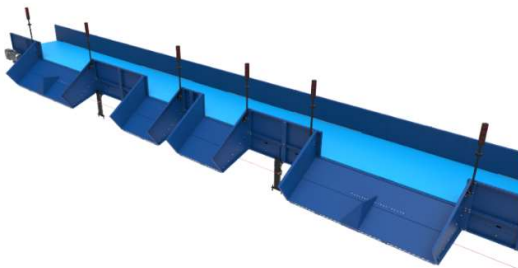


1. VDS Conveyor
2. VDS Loop Conveyor
3. VDS Chutes

### 9.2.1 VDS Conveyor:

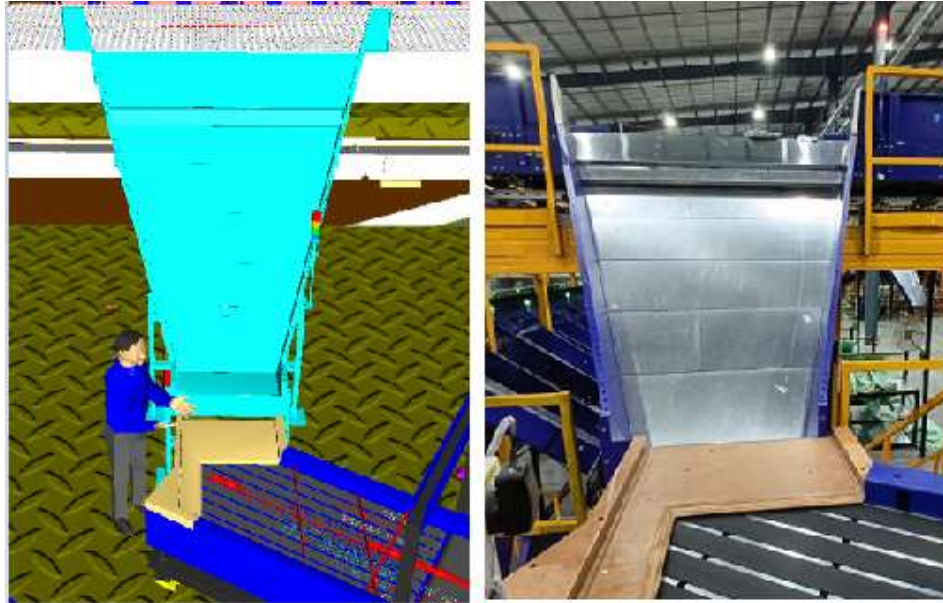


In this solution we are using ARB system for the VDS. Here ARB will divert/sort the shipment in to the VDS Chutes .





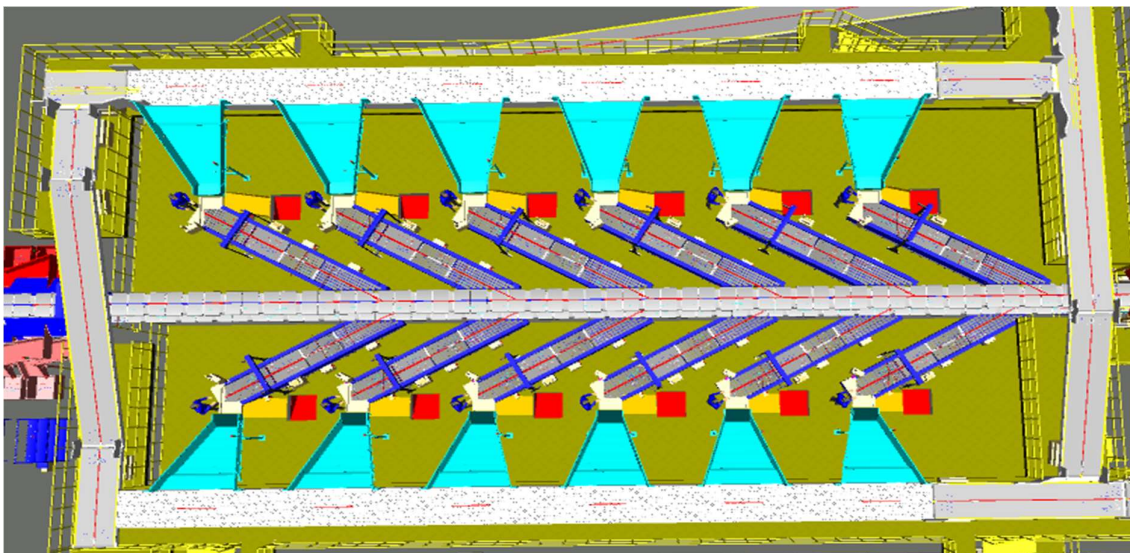
### 9.2.2 VDS Chutes



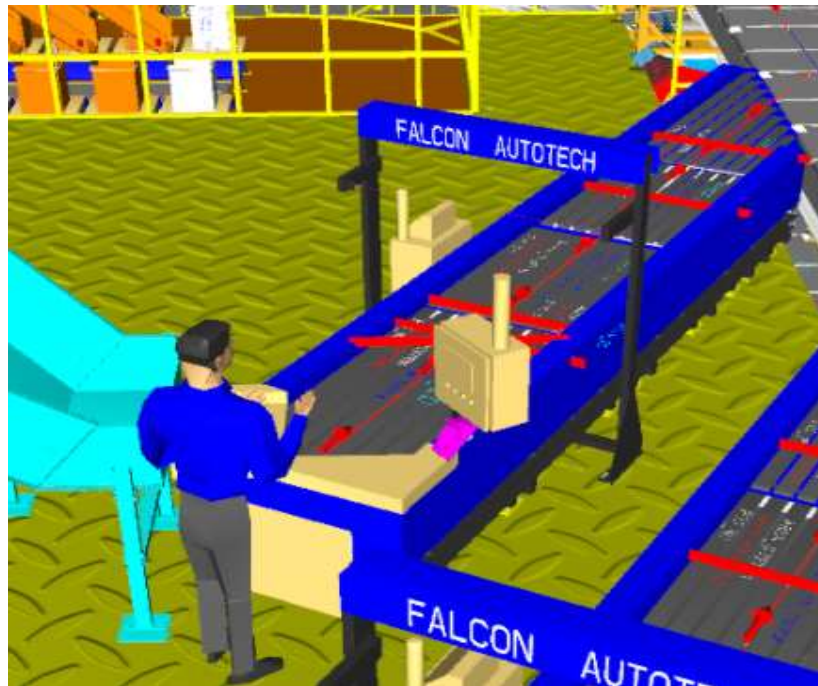
The material from the VDS Conveyor will divert on VDS Chute . From VDS Chute , Operators will pick up the parcel and feed them on feedline.

### 9.2.3 VDS Loop

Bulk materials will enter in to the VDS Loop through Infeed Highway Conveyors. Then the material will get distributed in to different VDS Chute maintaining load balance. The remaining material gets circulated in the VDS loop till the VDS Chute not be available for sorting .

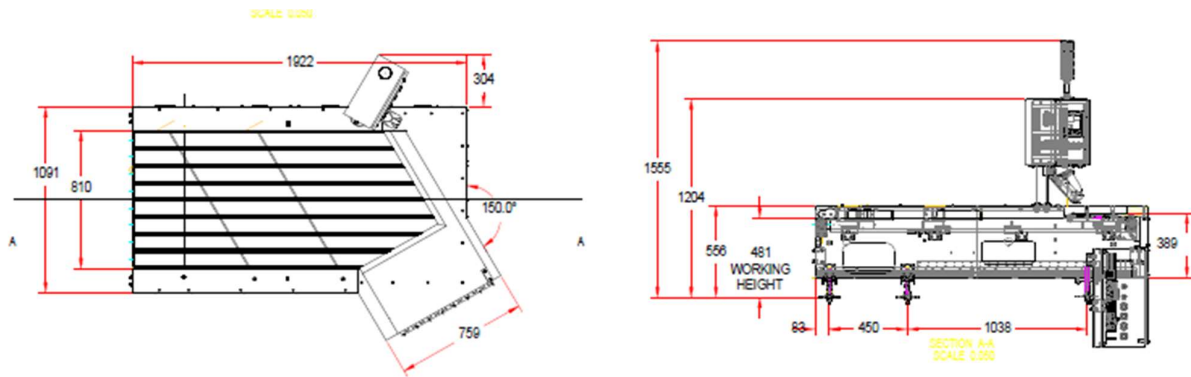
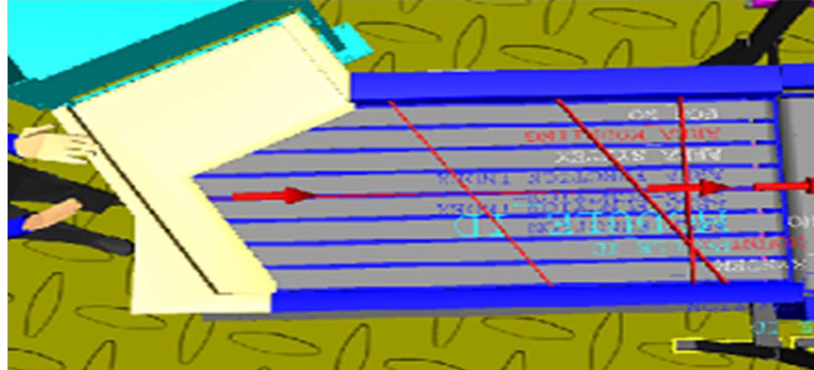


### 9.3 Feedlines:



1. Each Induct zone is equipped with 12 Induct lines ( 6 Nos on each Deck)
2. Each Feedline has four Conveyor Modules- 1. Loading & Orientation Conveyor, 1 weighing conveyor, 1 Intelligent feeding unit conveyor and 1 angle merge conveyor
3. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Parcel on the conveyor and prepare the Cross-Belt Cell for receiving the parcel with high precision.
4. Over the feedline, parcels are evenly spaced and smoothly transferred to the main Loop

### 9.3.1 Loading & Orientation Conveyor



Operator will pick the parcels from the VDS Chute and put them on Loading Conveyor

### 9.3.2 Weighing Conveyor

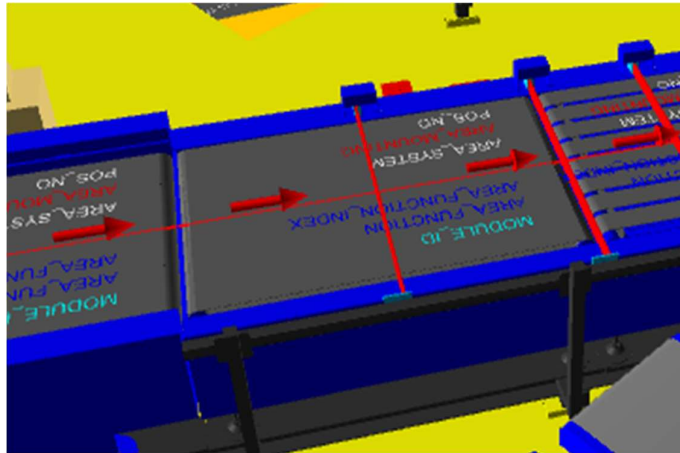
Weighing Conveyors equipped with high precision Load Cells to capture the weight of Parcels..

Make : (Bizerba/ Mettler Toledo)



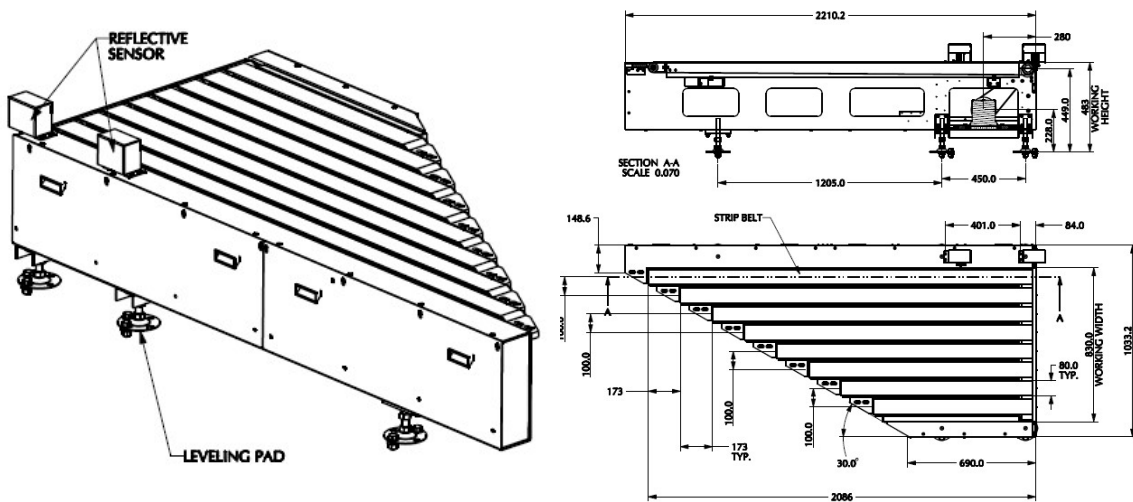
### 9.3.3 Spacer Conveyor

This conveyor is a variable speed special purpose module that creates space between parcels as well as regulates feeding on the intelligent merge.



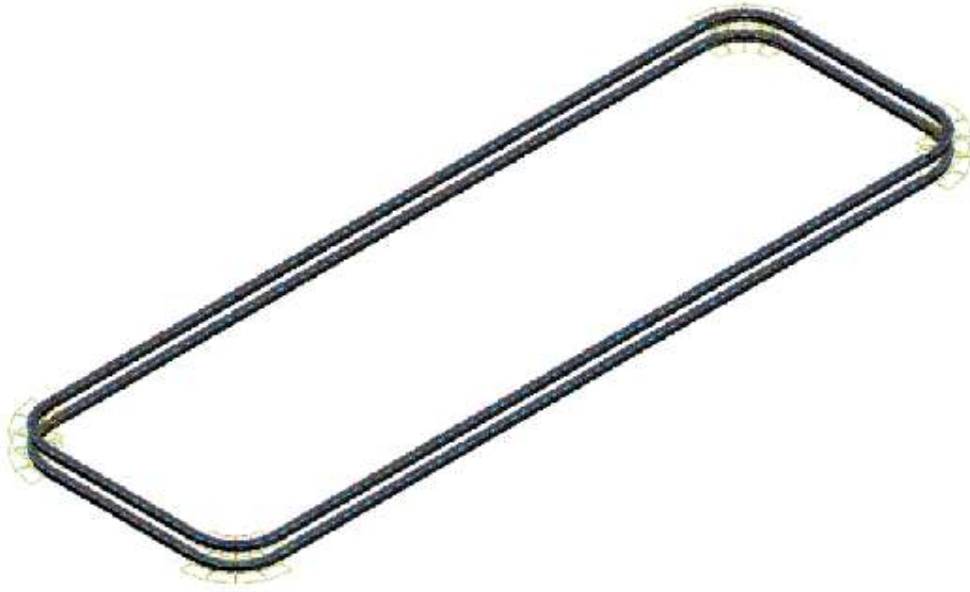
### 9.3.4 Intelligent Merge Conveyor

This is 30° triangular Dual Belt high-speed conveyor used for inducing parcel/boxes directly on to the sorter. The Belts are Strip Belts for smooth parcel movement.



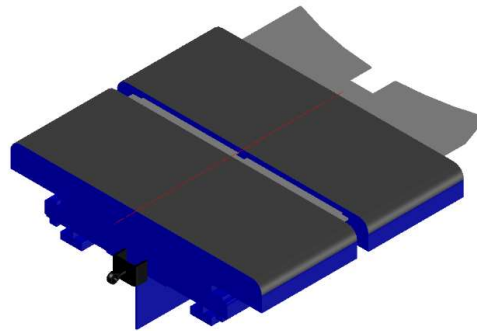


## 9.4 Main Loop



### Main Loop

1. The proposed Double Deck Sorter will be installed on the Mezzanine (3.6 metres from ground) with the Feedlines on the Mezzanine. The First Deck of Sorter will run at ~2800 mm from mezzanine floor at 2.1 m/s speed and second deck at ~4800 mm mezzanine floor at 2.1 m/s speed
2. We have provided Dual Belt Carriers with 1175 mm pitch.



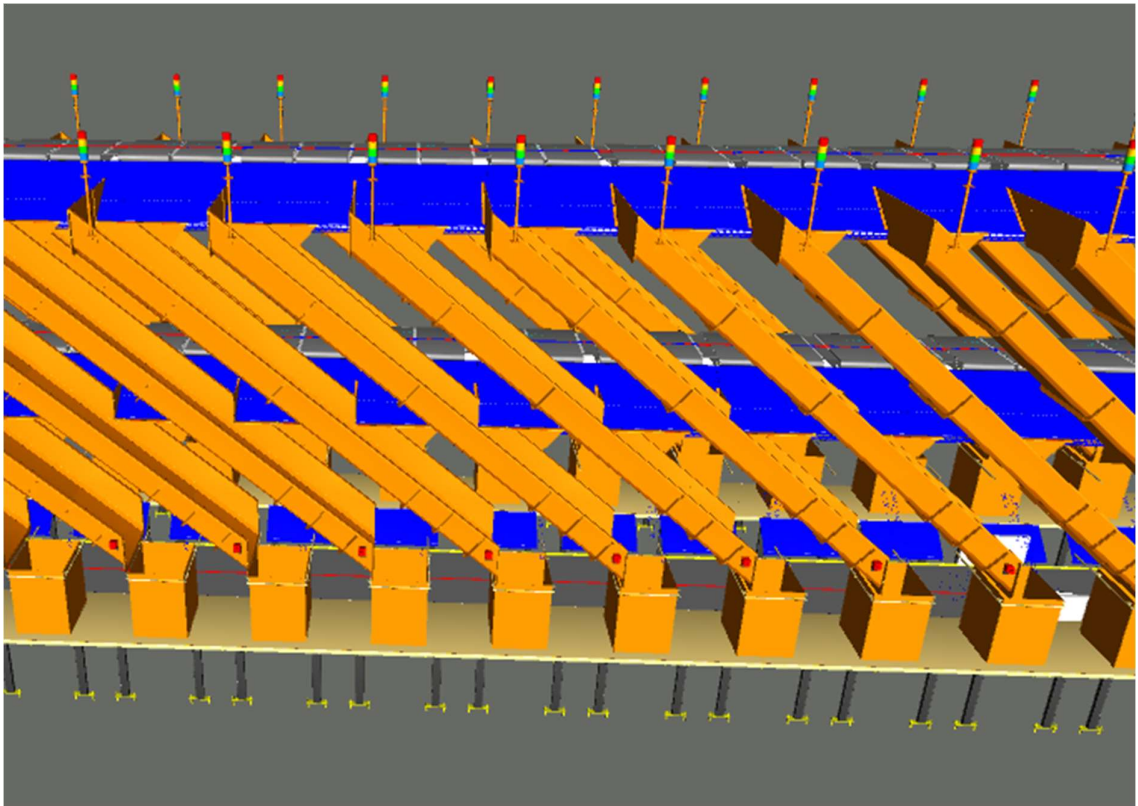
3. As Bags or Boxes pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the bags or boxes reaches respective chute, carrier carrying the shipment for the chute, will actuate and drop the bags or boxes in the assigned chute.

## 9.5 Output Chutes

Two Types of Output Chutes are used in this solution:

- Direct Bagging Chutes
- PTL Chutes

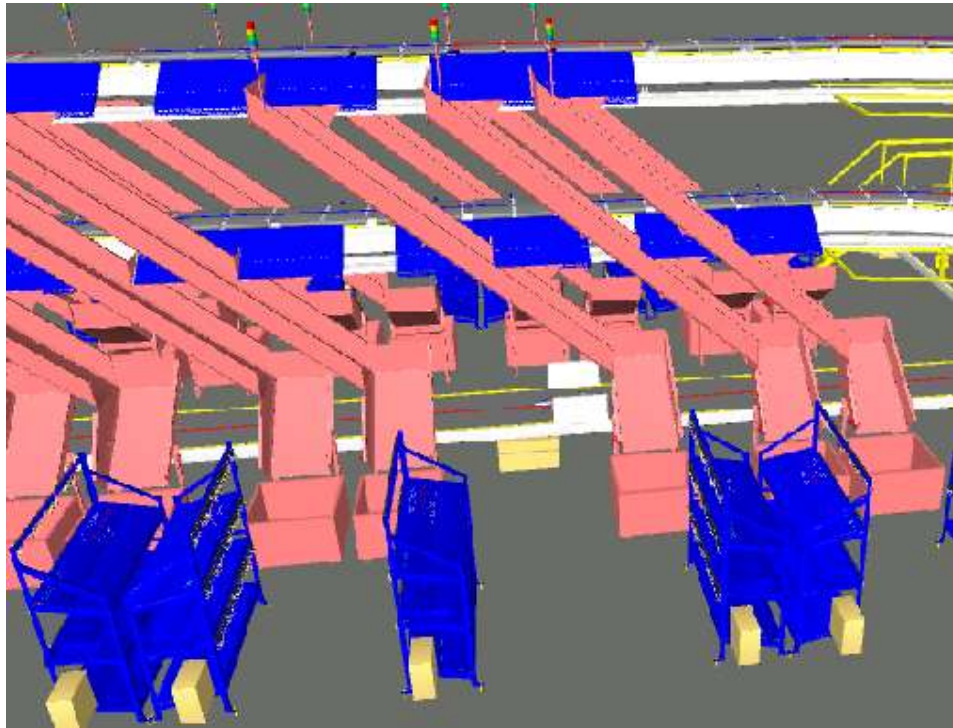
### 9.5.1 Direct Bagging Chutes



The sorted shipments will travel along these L-Shaped Sliding Chutes and get collected in the Bag holders. Upon fulfilment these bags will be closed and then pushed back onto the Takeaway Conveyors. These chutes are equipped with a shutter door, tower lamp, chute full sensor and push button.

| Chute IOs                       | Description                                                                                       |
|---------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Single Chute Full Sensor</b> | To alert the system that the chute is full and mark that chute unavailable for further sortation. |
| <b>Tower Lamp</b>               | To indicate the operator action on a chute                                                        |
| <b>Push Button</b>              | To Suspend/Resume Sorting into a particular Chute                                                 |

### 9.5.2 PTL Chutes



#### PTL Chutes

- These are collection type chutes where the parcels keep collecting into Bins.
- The Operators collect the parcels from Bin and will do secondary sorting by means of PTL Racks.

| Chute IOs                       | Description                                                                                       |
|---------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Single Chute Full Sensor</b> | To alert the system that the chute is full and mark that chute unavailable for further sortation. |
| <b>Tower Lamp</b>               | To indicate the operator action on a chute                                                        |
| <b>Push Button</b>              | To Suspend/Resume Sorting into a particular Chute                                                 |

### PTL Racks & Modules

- PTL Racks with PTL light modules has been provided for secondary sorting of parcels that have been dropped in the PTL chutes.
- Rack Configuration: 3 Rows, 4 Columns, total of 12 modules/rack in one chute.



- Operator will pick up a parcel from the PTL chute and scan the parcel barcode by handheld scanner
- The light of the block in the PTL rack which has been mapped to the destination location of the parcel will glow.
- Operator will place the parcel in the bag framed in that module and acknowledge by pressing the confirmation button on the module.
- This process will be repeated for all parcels falling in that chute bin.
- Once a bag is full, operator will take out the bag paste specific tag on bag seal and place it on the bag take-away conveyor.
- New bag will be placed in the empty module.



## 9.6 Steel Works

### 9.6.1 Mezzanine for CBS



Total ~5387 sq mtrs mezzanine platform required to install the CBS

### 9.6.2 Operator & Maintenance Platform



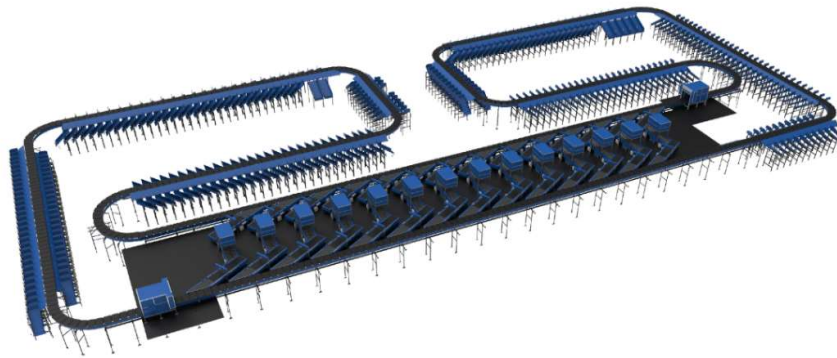
Total Area ~ 1500 sq metres of platform is required for operation & maintenance.

## 10. Description of Components

### 10.1 Cross Belt Sorter –

Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner. Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels.

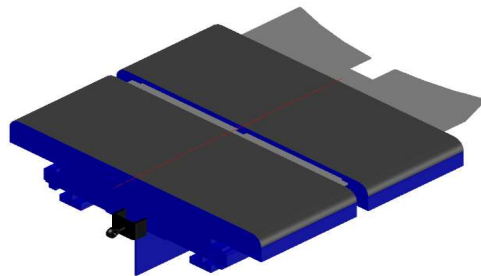
Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



**Cross Belt Sorter**

### 10.2 CBS Carrier

Falcon Autotech's CBS offers one of the highest Belt widths to Carrier Pitch Ratio in the market today. This additional belt width makes the system capable of handling larger product sizes without compromising on throughput. This also means reduced "dead area" between Carrier belts which drastically reduces the amount of "inbetweeners" and non-sortable parcel recirculation.



**CBS Carrier**

### 10.2.1 Servo Roller

- High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.



**Servo Roller**

### 10.2.2 Chassis

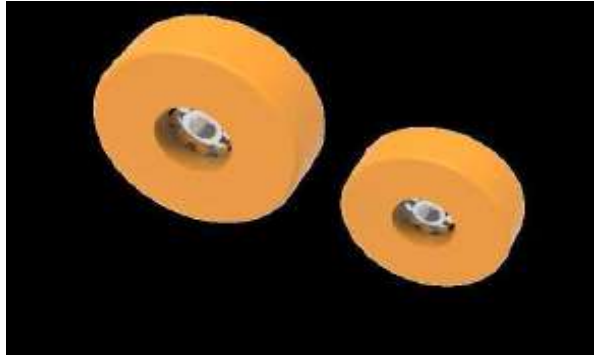
Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminum which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.



**Chassis**

### 10.2.3 Carrier Wheels

- Carrier wheels that are thoroughly tested and proved for long life cycle



### 10.2.4 Non-contact based linear motor drive

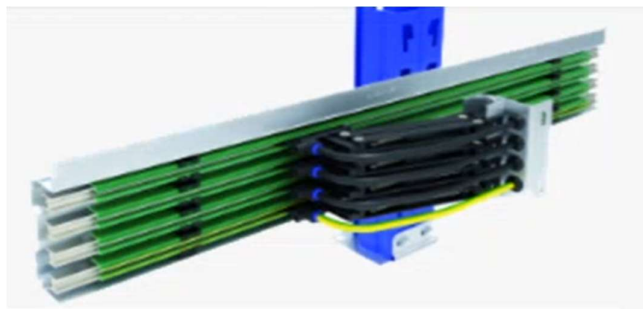
- Extremely High Energy Efficient Linear Induction Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed



**Linear Motor**

### 10.2.5 Power Transmission

- Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.

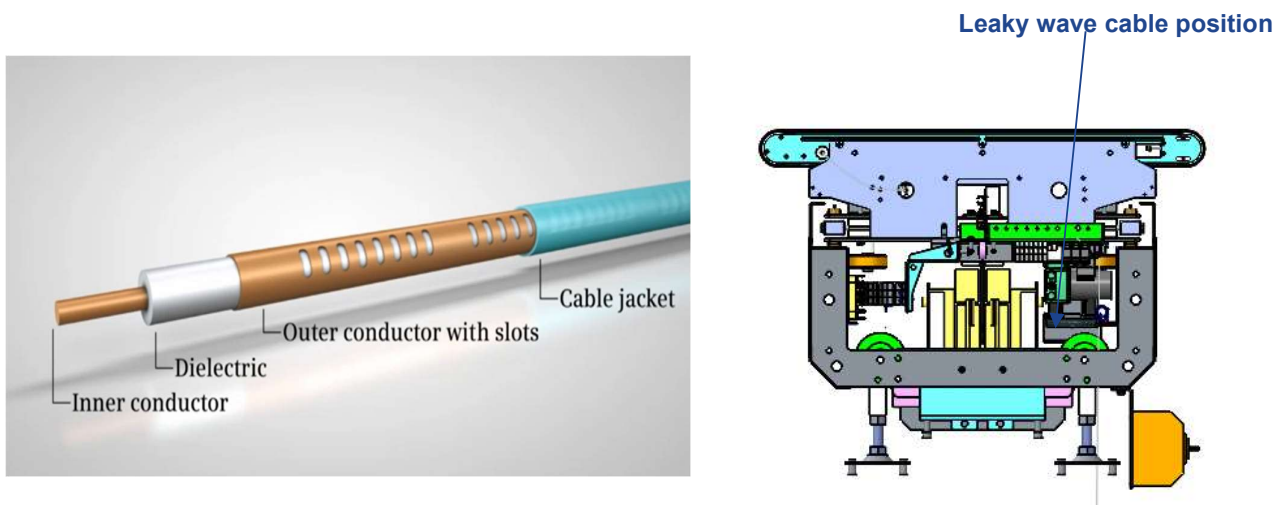




## Power Transmission

### 10.2.6 Data Transmission

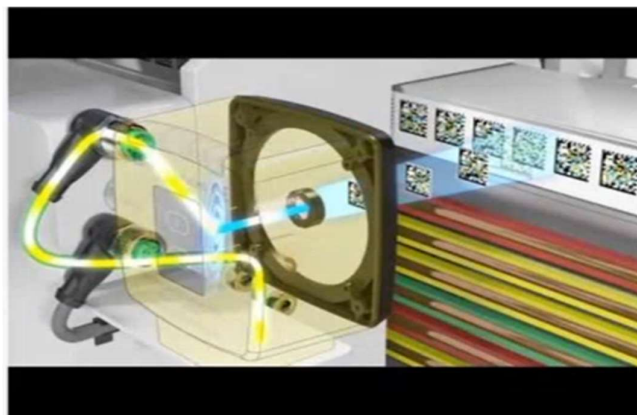
The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.



## Data Transmission

### 10.2.7 Carriers positioning System

Positioning system determines the exact location of each carrier in the loop at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter loop, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



## Carrier Positioning System

## 10.3 Scanning & Sensing on Main Sorter Loop

### 10.3.1 Automatic Barcode Scanning and Volume Scanning System

3 Side Barcode Scanner to scan parcel 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time. This will enable 5 sided Barcode Scanning. Additionally, the system is equipped with Volume Scanning System that can capture the Parcel dimensions with  $\pm 10$  mm accuracy.

Handheld Barcode Scanners are provided to run the machine on Semi- Automatic Mode, if required



### 10.3.2 Parcel Positioning System (Used for Physical & Virtual Centring)

Falcon Cross Belt Sorter is equipped with Multiple parcel positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Parcel on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).



### 10.3.3 Empty Carrier Detection System

This system detects the empty carriers on the sorter and prepares Feedlines to load the parcels on the empty carriers. (Camera based)



## 10.4 Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



| Specification           | UOM  | Remark                                                                                                        |
|-------------------------|------|---------------------------------------------------------------------------------------------------------------|
| Manufacturer Name       | Name | Falcon Autotech                                                                                               |
| Material of Roller      | Type | MS Rollers with Zinc Plating                                                                                  |
| MOC of Belt             | Type | PVC                                                                                                           |
| Belt Finish             | Type | Matt Finish                                                                                                   |
| Belt Joint Type         | Type | Vulcanized, Endless Belts                                                                                     |
| Belt Make/Model No.     | Make | FORBO/DERCO                                                                                                   |
| Motor Shaft             | MOC  | EN9 Shaft material,                                                                                           |
| Conveyor Under guarding | Spec | MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight         |
| Type of Guides          | Type | Steel guides at 50mm to 400mm height basis the position of conveyor on both sides from Conveyor Belt Surface. |
| Drive Power Rating      | kW   | Motor Rating depends on Length of the Conveyor                                                                |
| Type of Motor           | Type | AC Geared Motor                                                                                               |
| Type of Drive System    | Type | Direct Shaft Mounted Motor/ Flange Mounted                                                                    |
| Type of Drive           | Type | Chain drive/Torque Arm type                                                                                   |
| Drive                   | make | Lenze/Siemens/Omron/Allen Bradley                                                                             |
| Conveyor Speed          | m/s  | Variable speed to meet TPH requirements                                                                       |
| Type of Mounts          | Type | Fixed Heights Legs with Grouting Provisions                                                                   |

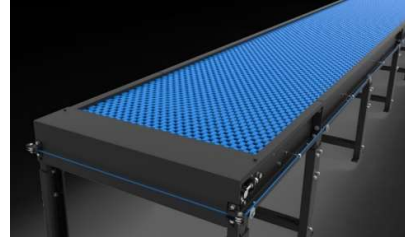


## 10.5 Plastic Modular Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

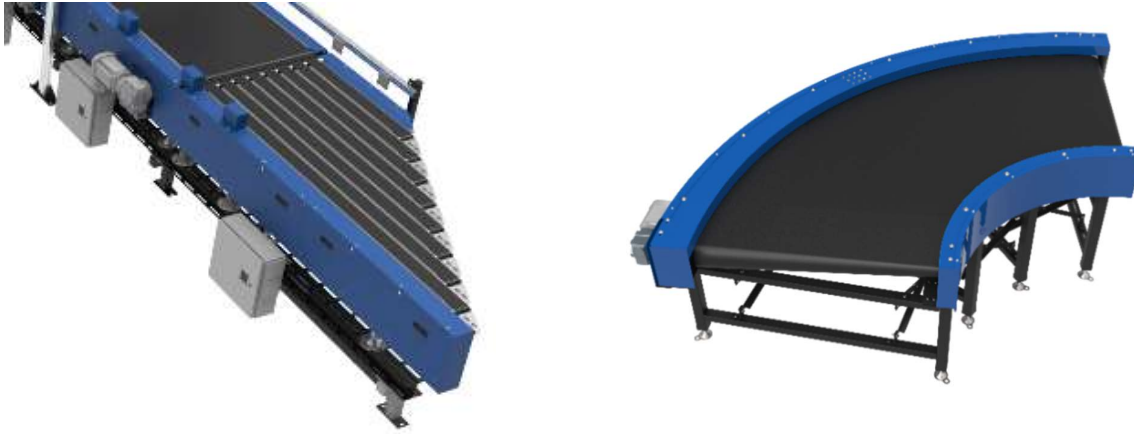
Some Salient Features of Falcon's Modular Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



| Specification                                            | UOM  | Remark                                                                                                     |
|----------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------|
| Manufacturer Name                                        | Name | Falcon Autotech                                                                                            |
| Plastic Modular Belt Material                            | Type | POM with PBT Pins, Pitch-25.4 mm, Pin Dia- 5mm                                                             |
| Plastic modular Belt Make/Model No.                      | Make | Movex/XP254 CR                                                                                             |
| Sprocket Size & No. of Sprockets on Drive End & Tail End | Type | 5/146.4 mm                                                                                                 |
| Sprocket material                                        | MOC  | POM                                                                                                        |
| Wear Strip Material                                      | MOC  | UHMW PE1000                                                                                                |
| Duplex Sprocket                                          | MOC  | MS                                                                                                         |
| Duplex Chain                                             | MOC  | MS                                                                                                         |
| Drive Shaft                                              | MOC  | EN 24                                                                                                      |
| Conveyor Under guarding                                  | Type | GI SHEET (1MM THICK)                                                                                       |
| Type of Guides                                           | Type | Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface. |

## 10.6 Powered Belt Merges & ZTR Curves (Pic for ref.)



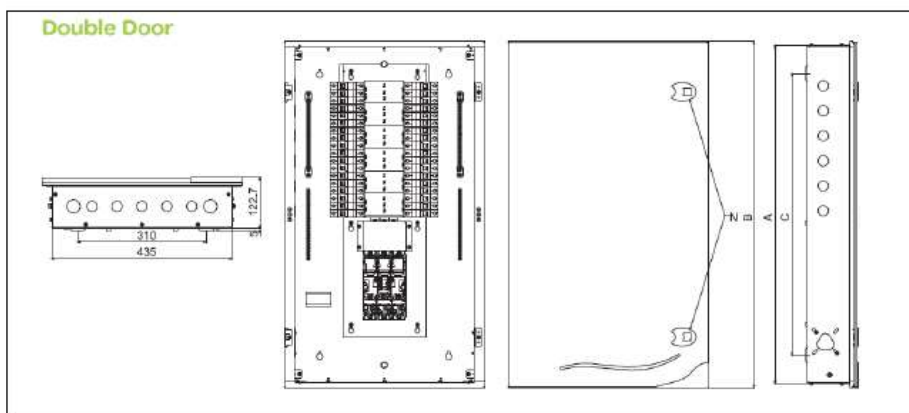
## 11. Electrical System

Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets.

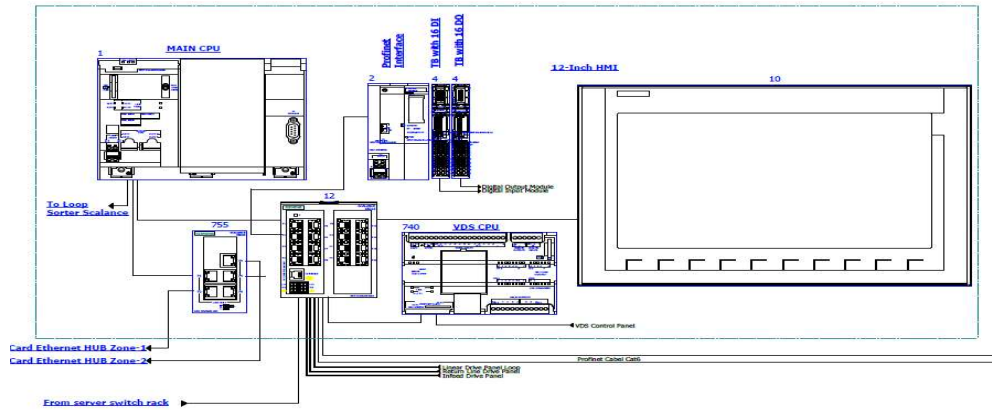
PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

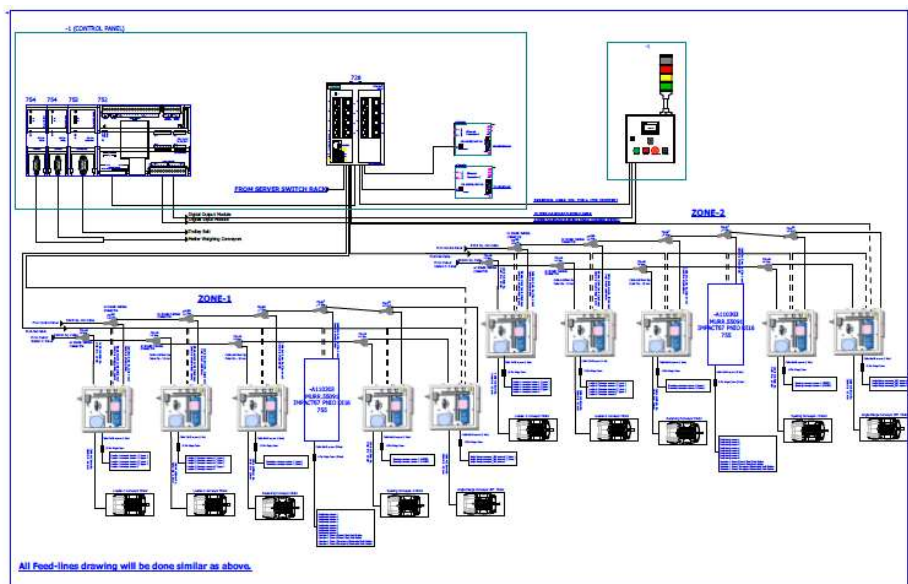
### 11.1 Reference Picture of Power Distribution Panel



### Main Control Panel (Reference)



### Induct Stations Control Panel (Reference)



## Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter.  
The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.  
All motors will have appropriate IP ratings

## Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

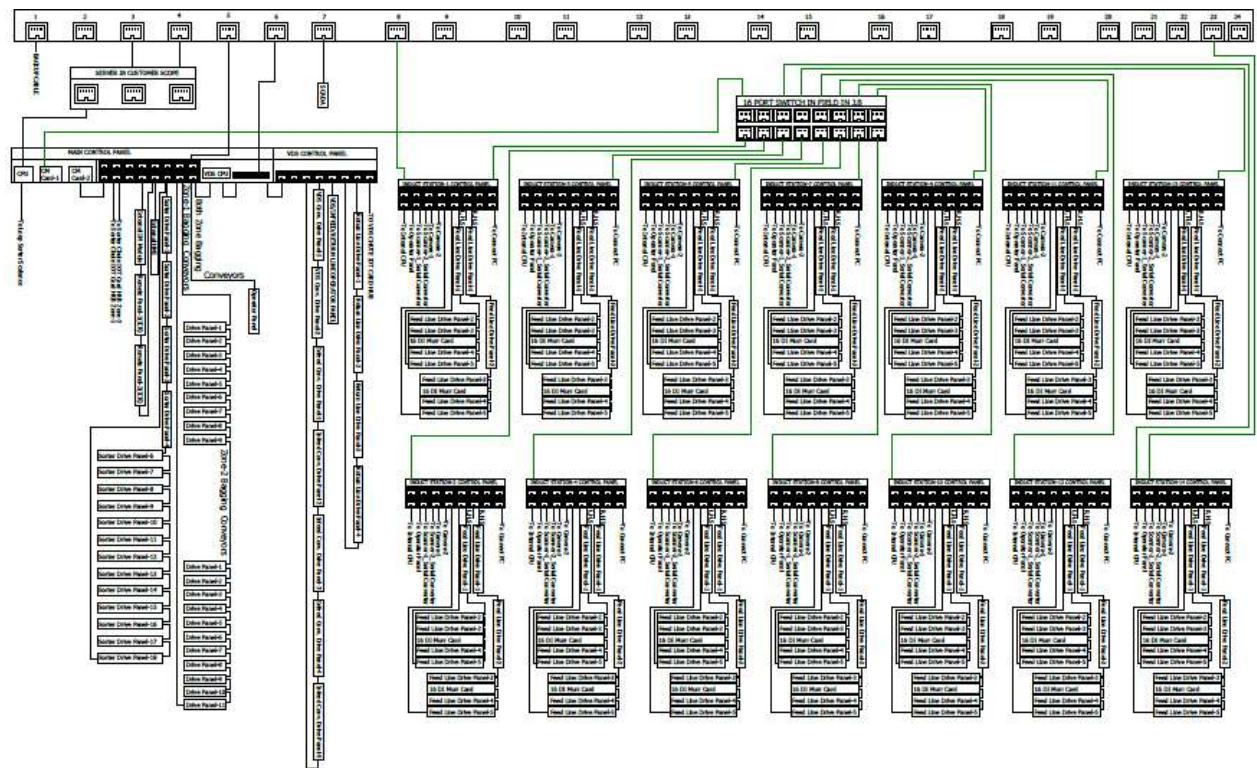
### Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform with optional Omron Platform for Conveyor Control. The entire system will be logically divided into zones (Sorter/ Feed Line/ Loop), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet + Ethernet + EtherCat

### Conveyor interface

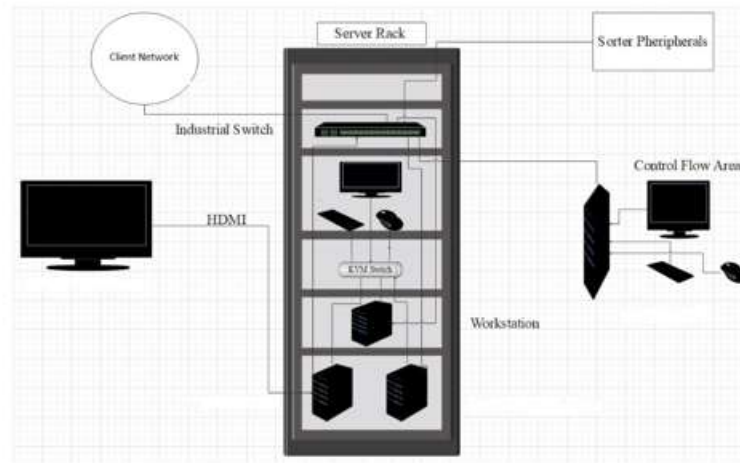
The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

## 12. Controls Architecture (Reference Only)





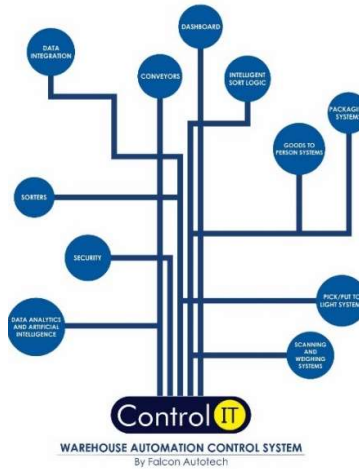
### 13. IT Architecture (Ref.)



## 14. High Availability Server Specifications

| Product/Model                | ITEM Description                                                                                                                                                                                                                                                                                                   | Qty. |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| T440 (4210R) Tower 2P Server | Intel Xeon Silver 4210R 2.4G, 10C/20T, 9.6GT/s, 13.75M Cache, Turbo, 4x16GB DDR4 (16 Slots), 2x1.2TB 10K RPM SAS 12Gbps 512n 3.5in Hot-Plug Hard Drive upto 8 Drives SAS/SATA, 2x1Gbe LOM, RAID 0,1,5 (H330), 2xRPS Standard Inbuilt 495W,iDRAC9,Enterprise, 3 Years Pro-Support & Mission Critical/Security Bezel | 2    |
| T40-Tower 1P SERVER          | Intel Xeon E-2224G Processor 3.5GHz 8M Cache,4C/4T,Turbo,71W,TPM, 2x8GB RAM (4 DIMM), 2x1TB SATA 3.5" 7.2k rpm HDD (3 Bay), 1x1Gbe LOM ,DVD Writer, Onboard RAID 01, Inbuilt PSU Standard (Max 1), 3 Year Onsite NBD/ Extra LAN Port, 480 GB SSD                                                                   | 1    |
| Synology_storage_DS720+      | 2 Bay Intel Celeron J4125 4-core 2.0 GHz, burst up, 2 GB DDR4 non-ECC Expand up to 7 drives with a DX517 expansion unit to 2.7 GHz,2 x built-in M.2 2280 NVMe SSD slots,3 Years                                                                                                                                    | 2    |
| SSD for Synology             | Seagate 1.92TB LE SSD                                                                                                                                                                                                                                                                                              | 4    |
| UPS                          | Microtek 1 KVA offline UPS                                                                                                                                                                                                                                                                                         | 3    |
| Dell Monitor                 | Dell 19.5" Monitor with HDMI And VGA Port                                                                                                                                                                                                                                                                          | 1    |
| KVM Switch                   | Aten KVM Switch (4 PORT)                                                                                                                                                                                                                                                                                           | 1    |
| License                      | Windows Server 2019 Standard R2 OEM                                                                                                                                                                                                                                                                                | 3    |
| Keyboard                     | Dell Keyboard Wired                                                                                                                                                                                                                                                                                                | 1    |
| Mouse                        | Dell Mouse Wired                                                                                                                                                                                                                                                                                                   | 1    |
| Cisco Switch                 | Cisco Business 350-24T-4G Managed Switch (CBS350-24T-4G-IN )                                                                                                                                                                                                                                                       | 1    |
| Converter                    | DP to VGA Converter Cadyce Brand                                                                                                                                                                                                                                                                                   | 1    |
| Server Rack                  | AC Server Rack                                                                                                                                                                                                                                                                                                     | 1    |
| 5 Meter LAN Cable            | 5 Meter D-link CAT6 Cable                                                                                                                                                                                                                                                                                          | 5    |

## 15. Falcon's WCS CONTROLIT



### Key Values

#### Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

#### Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

#### Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

#### Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

#### Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

## Supported Protocols

|             |                  |                         |                       |
|-------------|------------------|-------------------------|-----------------------|
| API         | FTP              | FILE UPLOAD / DASHBOARD | ODBC                  |
| WEBSERVICES | AMAZON S3 SERVER | FIREBASE                | Any Customer Protocol |

## 16. Proposed System Technical Details

### 16.1 Technical specifications of Sorter

| Items                                                        | Make                         |
|--------------------------------------------------------------|------------------------------|
| Sorter Carrier Type                                          | Dual Belt Regular            |
| Dual Belt Feature                                            | YES.                         |
| Sorter Speed in m/s                                          | Up to 2.1 m/s                |
| Sorter Loop Length (Lower Deck)                              | ~223.2 m                     |
| Sorter Loop Length (Upper Deck)                              | ~223.2 m                     |
| Length of the Sorter                                         | Layout Attached              |
| Width of the Sorter including Chutes                         | Layout Attached              |
| Sorter Actuation Technology                                  | Electric                     |
| Dual Belt Feature                                            | YES                          |
| Chutes + Rejection                                           | Layout Attached              |
| Total outputs Location                                       | Layout Attached              |
| Height of Lower Deck Sorter(from Mezzanine)                  | ~2800 mm                     |
| Height of Upper Deck Sorter(from Mezzanine)                  | ~4800 mm                     |
| Track Material & Type of Design to move the Carrier trolleys | ~Steel Track                 |
| Carrier Motor Make                                           | Falcon                       |
| Carrier Pitch                                                | 1175                         |
| No. of Carriers (Both Deck)                                  | ~380 Nos.                    |
| Motor/Drive type                                             | Linear Motor                 |
| Power Consumption                                            | Will be specified during DAP |
| Single Operator Feedlines                                    | 24                           |
| No. of Barcode Scanners (ICR type ).                         | 4                            |



| Power Transmission                            | Over Continuous Bus Bar |
|-----------------------------------------------|-------------------------|
| Parcel Positioning System                     | 4                       |
| Parcel Centering System                       | 4                       |
| Empty Carrier Detection System (Camera Based) | 4                       |
| Sorter Throughput (Both Deck)                 | ~32000                  |



## 16.2 Mechanical Equipment- Bill of Materials

| Pos. | Qty. | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Value                                                                              |
|------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| FM1  | 1    | <b>Conveyor Package</b> <ul style="list-style-type: none"> <li>Powered Belt Conveyors</li> <li>Powered Turn Conveyors (90 Degrees)</li> <li>ARB Conveyors</li> <li>ARB Diverter</li> <li>Telescopic Belt Conveyor (5+10)</li> </ul>                                                                                                                                                                                                                                                                                                                                 | ~1039 Metres (86 Conveyors)<br>2 Nos.<br>~100 Metres (4 Nos).<br>24 Nos<br>17 Nos. |
| FM2  | 24   | <b>Feedlines</b><br>Consists of <ul style="list-style-type: none"> <li>Single Operator Loading Conveyor.</li> <li>Weighing System</li> <li>Dynamic Spacing System</li> <li>Angle merge</li> <li>Hand Scanners for Manual Scanning</li> </ul>                                                                                                                                                                                                                                                                                                                        | 1 Set<br>1 Set<br>1 Set<br>1 Set<br>1 No.                                          |
| FM4  | 1    | <b>Falcon Lower Deck Sorter</b><br><b>1 Loop Cross Belt Sorter</b> <ul style="list-style-type: none"> <li>Sorter Height</li> <li>Sorter Length</li> <li>Sorter Speed</li> <li>Sorter Drive</li> <li>Carrier Pitch</li> </ul> Including: <ul style="list-style-type: none"> <li>Standard Sorter Supports</li> <li>Empty Carrier Detection System</li> <li>Product Centring System</li> <li>Camera based Barcode reader</li> <li>Dimension Scanning System</li> <li>E-Stops</li> <li>Hooters</li> <li>Fencing</li> <li>Netting</li> <li>Overshoot Assembly</li> </ul> | 2800 mm<br>~223 m<br>~2.1 m/s<br>Linear Motor<br>1175 mm                           |

|     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                 |
|-----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| FM5 | 1 | <b>Falcon Upper Deck Sorter</b><br><b>1 Loop Cross Belt Sorter</b> <ul style="list-style-type: none"> <li>• Sorter Height</li> <li>• Sorter Length</li> <li>• Sorter Speed</li> <li>• Sorter Drive</li> <li>• Carrier Pitch</li> </ul> Including: <ul style="list-style-type: none"> <li>• Standard Sorter Supports</li> <li>• Empty Carrier Detection System</li> <li>• Product Centring System</li> <li>• Camera based Barcode reader</li> <li>• Dimension Scanning System</li> <li>• E-Stops</li> <li>• Hooters</li> <li>• Fencing</li> <li>• Netting</li> <li>• Overshoot Assembly</li> </ul> | 4800 mm<br>~223 m<br>~2.1 m/s<br>Linear Motor<br>1175 mm                        |
| FM6 | 1 | <b>Chutes</b> <ul style="list-style-type: none"> <li>• VDS Chutes</li> <li>• PTL Chutes</li> <li>• L Type Collection Chute</li> <li>• Rejection Chutes</li> <li>• Irregular Chutes</li> <li>• Chute Full Sensors</li> <li>• Tower Lights</li> <li>• Bag Holder Assembly</li> </ul>                                                                                                                                                                                                                                                                                                                | 24 Nos<br>54 Nos<br>100 Nos<br>6 Nos<br>26 Nos<br>184 Nos<br>184 Nos<br>100 Nos |
| FM7 | 1 | <b>PTL</b> <ul style="list-style-type: none"> <li>• PTL Racks</li> <li>• PTL Modules</li> <li>• PTL Control Box</li> <li>• PTL Barcode Hand scanner 1-D</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                | 808 Nos<br>808 Nos<br>69 Nos<br>68 Nos                                          |
| FM8 | 1 | <b>Steel Works</b> <ul style="list-style-type: none"> <li>• Operator and Maintenance Platform</li> </ul> <p><b><u>Please note- The current Proposal does not include the Price of Steel Works and the same shall be discussed separately with yourself.</u></b></p>                                                                                                                                                                                                                                                                                                                               | ~6887 SQM                                                                       |

### 16.3 Electrical Equipment

| Electricals |   |                                                                                                                                                                                                                                                                                                                                                  |          |
|-------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| FC1         | 1 | <b>Consists of</b> <ul style="list-style-type: none"> <li>• Main Power Distribution panels</li> <li>• Main Control Panels</li> <li>• Feedline Control Panels</li> <li>• Conveyor Drive Panels</li> <li>• Sorter Drive Panels</li> <li>• Network Switches</li> <li>• Field Cabling</li> <li>• Cable trays</li> <li>• Remote I/O Panels</li> </ul> | Included |

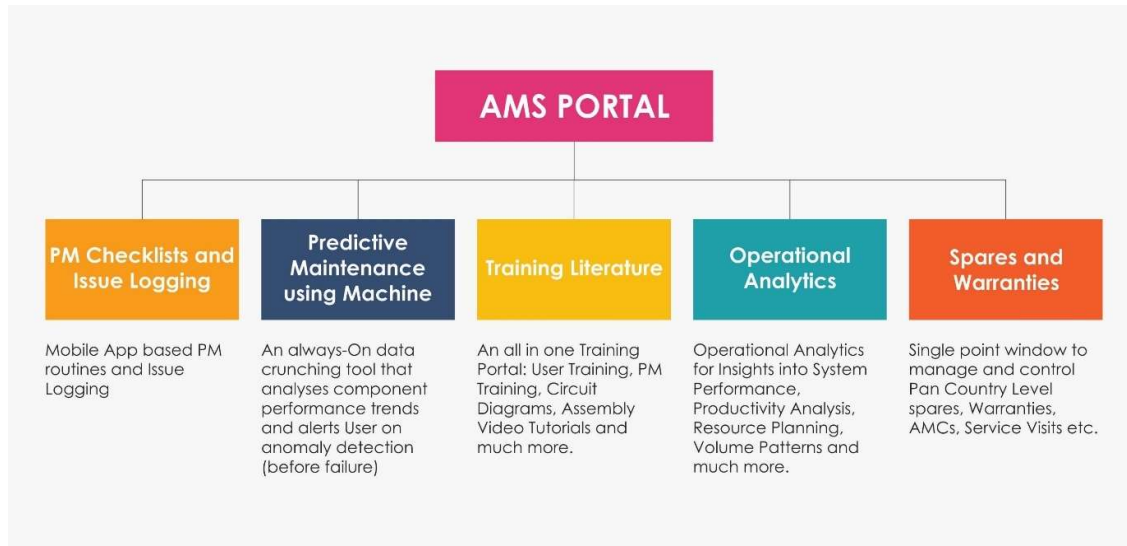
### 16.4 Control System

| Components |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
|------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| FC2        | 1 | <b>Controls and IT Infrastructure Controls</b> <ul style="list-style-type: none"> <li>• Weigh Scale Control</li> <li>• Barcode Scanner Controls</li> <li>• Dimension Scanner Controls</li> <li>• Conveyor Package Controls</li> <li>• Sorter Control</li> <li>• Induct Control</li> <li>• Smart Chutes Control</li> </ul><br><b>IT</b> <ul style="list-style-type: none"> <li>• Control IT Software Package</li> <li>• High Availability Server Setup (Microsoft Clustering based)</li> <li>• Networking Switches</li> </ul> | Included |

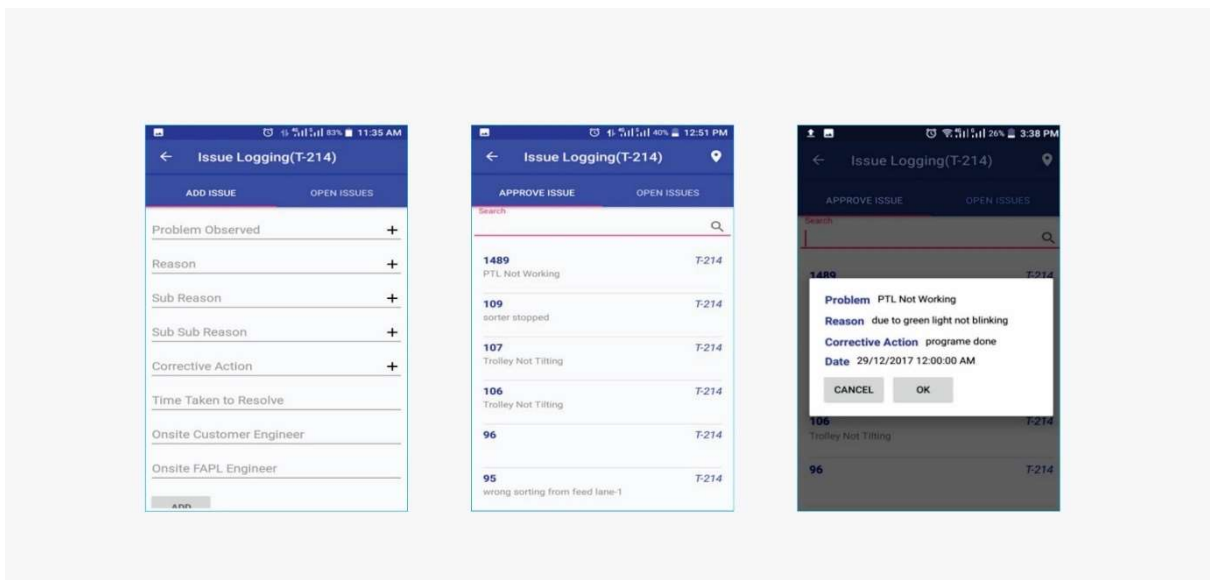
## 17 Key Components Make

| Items                      | Make                                                        |
|----------------------------|-------------------------------------------------------------|
| PVC Belts                  | Forbo/Derco                                                 |
| Rollers                    | Falcon                                                      |
| TBCs                       | Wangman/ Guanchao/ Pierski                                  |
| Cross Belt Carriers        | Falcon                                                      |
| Linear Induction Motors    | Falcon                                                      |
| Feed Line Motors           | Servo Motors: Lenze,<br>Induction Motors – SEW/ Lenze/ Nord |
| Barcode Readers            | SICK                                                        |
| Wireless Barcode Scanner   | Honeywell/Motorola                                          |
| Weighing Conveyors         | Bizerba / Mettler Toledo                                    |
| Volume Scanners            | SICK                                                        |
| Sensors                    | Sick/Leuze/ P&F                                             |
| PLC                        | Siemens + Omron                                             |
| Control Panels             | Rittal/ BCH                                                 |
| MDR Rollers                | Falcon/ Pulse/ Interrol                                     |
| Power Transmission Systems | Vahle                                                       |

## 18 Automation Management System (AMS)

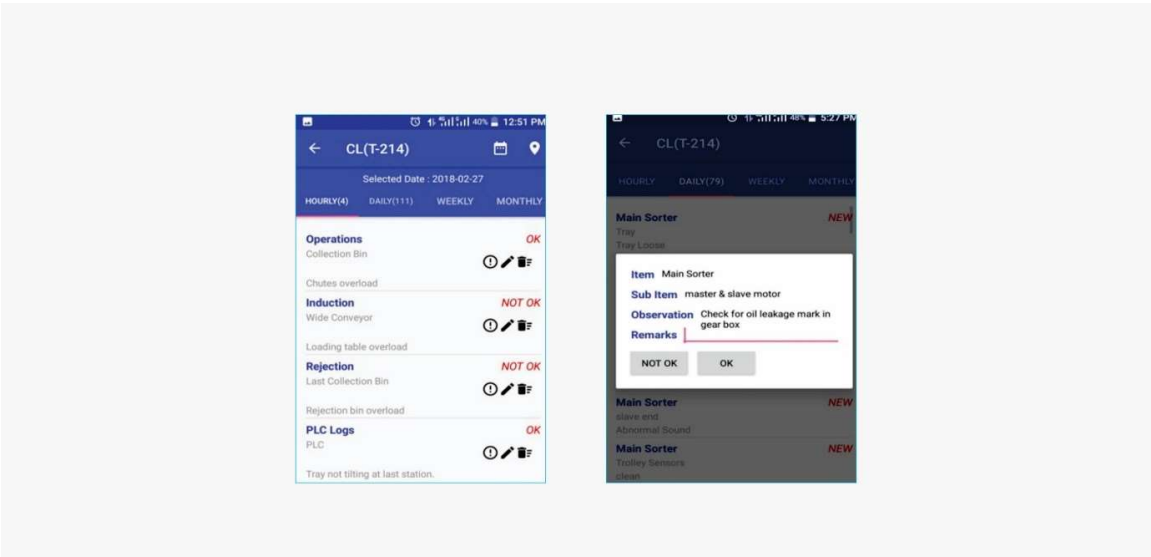


### a. Issue Logging

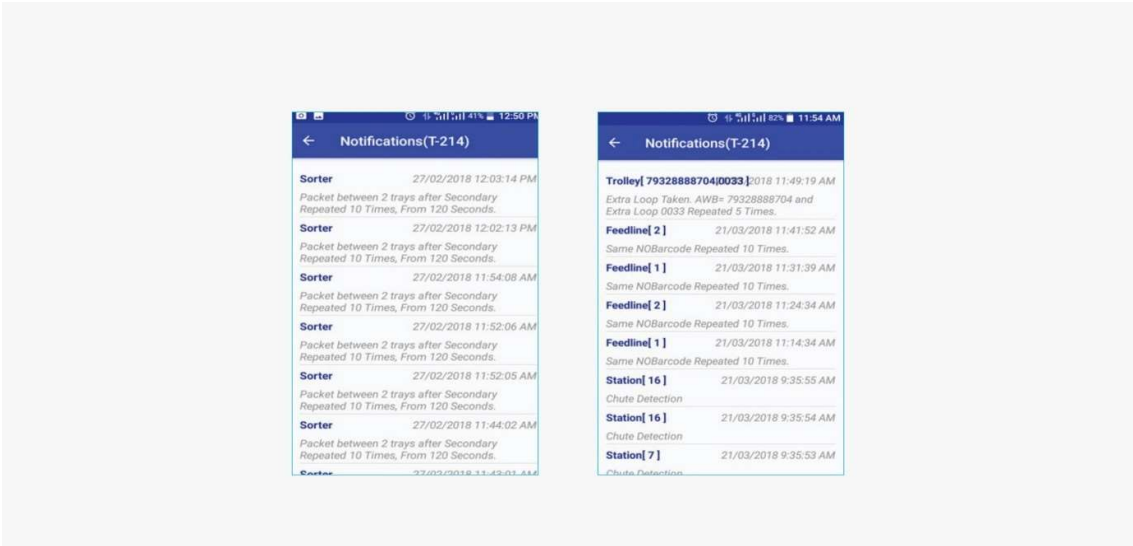




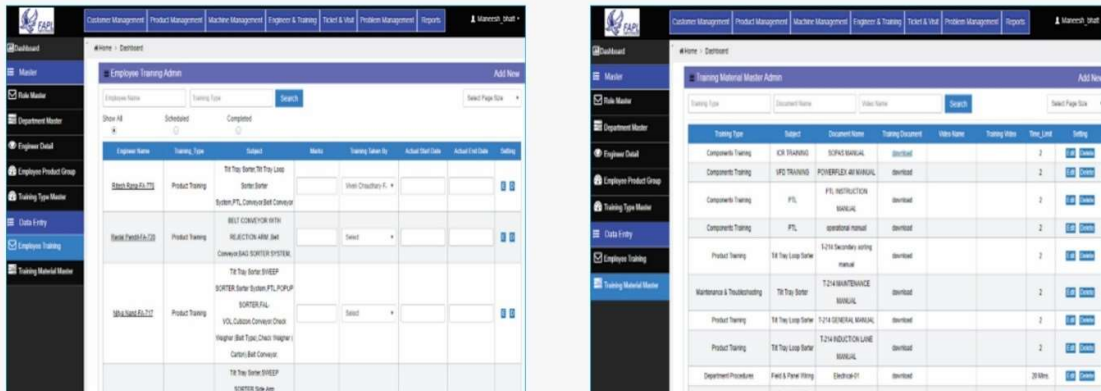
b. Preventive Maintenance Check List



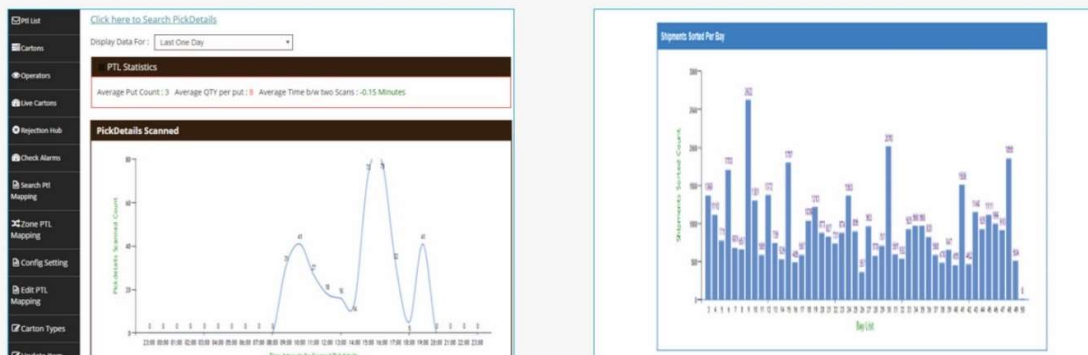
c. Predictive Maintenance



#### d. Training Portal



### e. Operational Analysis



## 19 Commercials

### 19.1 Price List

| S.NO                                                               | Package                   | Price (INR)    |
|--------------------------------------------------------------------|---------------------------|----------------|
| 1                                                                  | Conveyors Package         | ₹ 10,84,27,261 |
| 2                                                                  | Cross Belt Sorter Package | ₹ 15,70,83,585 |
| 3                                                                  | Destinations Package      | ₹ 2,89,24,550  |
| 4                                                                  | Service Package           | ₹ 2,28,74,520  |
| <b>Total</b>                                                       |                           | ₹ 31,73,09,916 |
| <b>"Business Cooperation Agreement"<br/>Discount for Delhivery</b> |                           | 4.50%          |
| <b>Final Total</b>                                                 |                           | ₹ 30,30,30,970 |

### 19.2 Breakup

#### 19.2.1 Conveyors Package

| S.NO         | Product Group                                                                                                                                         | Units | Qty  | Total Price (INR)     |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|-----------------------|
| 1            | PVC Belt Conveyor                                                                                                                                     | mt    | 1039 | ₹ 4,66,36,863         |
| 2            | PVC Conveyor Drive Units                                                                                                                              | Count | 86   |                       |
| 3            | ARB Conveyor                                                                                                                                          | mt    | 100  | ₹ 2,45,28,000         |
| 4            | ARB Diverter                                                                                                                                          | Count | 24   |                       |
| 5            | Telescopic Belt Conveyors<br>Model (5+10)                                                                                                             | Count | 17   | ₹ 2,66,31,669         |
| 6            | 90 Degree Curve Conveyor- Modular (1200 mm<br>Width)                                                                                                  | Count | 2    | ₹ 15,40,000           |
| 7            | <b>Control Package-</b><br>Wiring +<br>Cable Trays+<br>E-Stops and Pull Chords +<br>PLC Control panels with HMI+<br>Tower Lamps and Hooters+<br>SCADA | Count |      | ₹ 90,90,729           |
| <b>Total</b> |                                                                                                                                                       |       |      | <b>₹ 10,84,27,261</b> |

**19.2.2 Cross belt Sorter package**

| S.NO | Product Group                                                                                                                                                                                                                                                                                                                                                                   | Units | Qty | Total Price (INR)     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----------------------|
| 1    | <b>Lower Deck CBS-</b><br><br>Loop Length- 212.4 meters<br>Number of Carriers- 181<br>Number of Belts- 362<br>2 X ICR & VMS<br>2 X ECDS System<br>2 X Centring System<br>2 X Positioning System<br>1 X Loop Control System<br>1 X Power Distribution System<br>1 X Carrier Positioning System<br>1 X Data Transmission System<br>Drive System<br>Sorter Control System with HMI | Count | 1   | ₹ 5,83,02,791         |
| 2    | <b>Upper Deck CBS-</b><br><br>Loop Length- 212.4 meters<br>Number of Carriers- 181<br>Number of Belts- 362<br>2 X ICR & VMS<br>2 X ECDS System<br>2 X Centring System<br>2 X Positioning System<br>1 X Loop Control System<br>1 X Power Distribution System<br>1 X Carrier Positioning System<br>1 X Data Transmission System<br>Drive System<br>Sorter Control System with HMI | Count | 1   | ₹ 5,66,64,662         |
| 3    | <b>Feedlines Comprising of-</b><br><br>Loading Conveyors<br>Spacing Conveyors<br>MLG System<br>Merge Conveyors<br>Weighing Conveyor<br>Control System with HMI                                                                                                                                                                                                                  | Count | 24  | ₹ 4,21,16,132         |
|      | <b>Total</b>                                                                                                                                                                                                                                                                                                                                                                    |       |     | <b>₹ 15,70,83,585</b> |

**19.2.3 Destination Package**

| S.NO                     | Chute Type                       | Qty | Price (INR)    |
|--------------------------|----------------------------------|-----|----------------|
| 1                        | Direct Bagging Chutes            | 100 | ₹ 1,30,74,070  |
| 2                        | PTL Chutes                       | 54  |                |
| 3                        | Rejection Chutes                 | 6   |                |
| 4                        | VDS Chutes                       | 24  |                |
| 5                        | Irregular Shipment Chutes Type 1 | 24  |                |
| 6                        | Irregular Shipment Chutes Type 2 | 2   |                |
| 7                        | Irregular Transfer Chute         | 24  |                |
| Bag Holders & Chutes I/O |                                  |     |                |
| 8                        | Chute Full Sensors               | 184 | ₹ 65,96,000.00 |
| 9                        | Tower lamps (2 colour)           | 184 |                |
| 10                       | Push Button                      | 184 |                |
| 11                       | I/O for Chute full sensors       | 139 |                |
| 12                       | Bag Holder                       | 100 |                |
| PTL                      |                                  |     |                |
| 13                       | PTL Racks Bagging                | 808 | ₹ 92,54,480.00 |
| 14                       | PTL Modules                      | 808 |                |
| 15                       | PTL Control Box                  | 59  |                |
| 16                       | PTL hand barcode scanner 1D      | 68  |                |
|                          | Total                            |     | ₹ 2,89,24,550  |

*\*Destination Package also includes the costing of manual PTL setup*



**19.2.4 Services Package**

| S.NO         | Service                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Qty | Price (INR)          |
|--------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------------|
| 1            | Packaging and Forwarding       | Wooden/Cardboard packing of system modules along with associated documentation and labelling for a safe transport to Client Site.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1   | ₹ 58,90,304          |
| 2            | Project Management             | <p>A Dedicated Project Delivery group will be earmarked for the project comprising of</p> <ul style="list-style-type: none"> <li>- 1 X Project Engineer</li> <li>- 1 X Project Manager</li> <li>- 1 X Team leader</li> </ul> <p>This group will be supervised by Head of Projects and shall work very closely with your teams for a smooth execution of project and any risk mitigation.</p> <p>This also included preparation of Detailed Engineering Drawings, Layouts, Business requirement Documents, Manuals and Guides.</p> <p>This also includes travel to the site and your Head office on need basis</p> <p>This also includes coordinating with your various vendors, consultants, contractors etc</p> | 1   | ₹ 15,78,656          |
| 3            | Installation and Commissioning | <p>This includes the complete mechanical installation, electrical installation, Commissioning, Training, stabilization support and handover of the system to the client.</p> <p>During this phase, a team comprising of Site Manager + Electrical Engineers + Mechanical Engineers + PLC Engineers + Software Engineers +Store Manager + Safety In charge will be stationed at your site and will supervise the satisfactory delivery of the project. We shall be hiring a reputed manpower contractor for executing this project.</p> <p>The Interstate Travel, Local travel, Stay, Food, daily allowances are all included in this</p>                                                                         | 1   | ₹ 1,47,25,760        |
| 4            | Software Services              | This includes the Supply of Falcon's Sort IT WCS Package customized to your requirement + Custom IT integration with your Host IT Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1   | ₹ 6,00,000           |
| 5            | Tool Kit                       | This includes the basic tools required for the maintenance of System                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1   | ₹ 79,800             |
| <b>Total</b> |                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     | <b>₹ 2,28,74,520</b> |

The Total Price is to be understood

- Fright: Extra
- Including Packing and Forwarding
- Duty and Taxes Extra
- Installation and Commissioning: Mentioned above.

\*Material Unloading & Laydown area is in customer scope.

\*Scaffoldings, Man Lift etc. required for reaching the maintenance platforms shall be in customer scope

## 20 Payment Terms

Stage 1: 40% Advance along with Purchase Order

Stage 2: 50% along with 100% Taxes against Delivery of Material on Site on Pro-rata basis.

Stage 3: 10% after successful installation and commissioning.

## 21 Warranty Period

Falcon offered Systems come with a standard warranty of 12 months from the date of commissioning of the system or 15 months from the date of completion of dispatch of the materials whichever is earlier. End of commissioning is considered as the date on which the system is taken into operation for regular sorting activity. In case of delay in site readiness or lifting of materials from Falcon works, the 15-month period will be considered to start from the date of official communication of material readiness at Falcon end. The warranty include:

- 24 X 7 Telephonic, Email and Remote Service Support when required
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
  - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
  - Negligence or abnormal use of equipment.
  - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
  - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

## 22 Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Scaffoldings, Man Lift etc. required for reaching the maintenance platforms.
- Air Compressor
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- TV Screens for Docks and Chutes
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.